

Think about the water you see and use every day. It falls from the sky as rain; it fills the rivers and oceans. You drink it, you cook with it, you bathe in it and you wash your clothes in it. Reactions in your body take place between chemicals that are dissolved in it. Without water, there is no life.

In this chapter, you will learn what makes water crucial for life. You will learn about the sources of water here on Earth and how water sampling and testing can be undertaken to check water quality.

You will learn about some of the unique properties of water and the way in which these properties support life on Earth. Lastly, you will also learn how each property can be explained by the structure and bonding present in water.

### Science as a human endeavour

- the supply of potable drinking water is an extremely important issue for both Australia and countries in the Asian region. Water sourced from groundwater and seawater undergoes a number of purification and treatment processes (such as desalination, chlorination, fluoridation) before it is delivered into the supply system. Chemists regularly monitor drinking water quality to ensure that it meets the regulations for safe levels of solutes. Heavy metal contamination in ground water is monitored to ensure that concentrations are at acceptable levels. Several methods can be used to reduce heavy metal contamination; the method used is influenced by economic and social factors.

### Science understanding

- the shape and polarity of molecules can be used to explain and predict the nature and strength of intermolecular forces, including dispersion forces, dipole–dipole forces and hydrogen bonding
- the unique physical properties of water, including melting point, boiling point, density in solid and liquid phases, and surface tension, can be explained by its molecular shape and hydrogen bonding between molecules
- the solubility of substances in water, including ionic and polar and non-polar molecular substances, can be explained by the intermolecular forces, including ion–dipole interactions between species in the substances and water molecules, and is affected by changes in temperature

## 15.1 Essential water

Water is a special chemical. It moderates our weather, shapes our lands and is essential for the existence of life. Water is the most abundant liquid on Earth, covering more than 70% of our planet. The total amount of water is estimated to be more than 1.3 billion cubic kilometres.

This water on Earth constantly cycles, as shown in Figure 15.1.1. It is in continuous movement between land, ocean, rivers and creeks and the atmosphere. Warmth from the Sun causes the water in oceans, lakes and rivers to evaporate and become vapour in the air. As water vapour rises, the air temperature drops and the water vapour condenses to form droplets, which merge to form clouds. When the water droplets are large enough, water falls back to Earth as rain, ice or snow, depending on the air temperature.

Other processes also add water to the atmosphere. When fossil fuels undergo combustion, water is a product and is released into the air. Water also evaporates from the leaves, stems and flowers of plants.

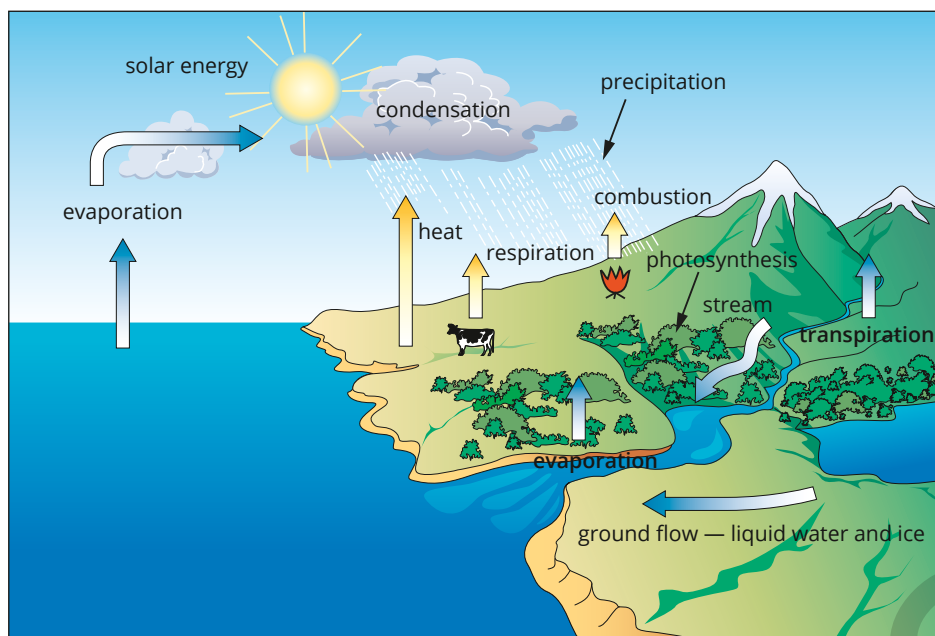


**FIGURE 15.1.1** The water cycle. The arrows show some of the ways water on Earth is redistributed.

Water is an exceptional solvent for a large variety of substances. However, the solvent properties of water do cause problems. Because water dissolves many chemicals readily, it can be easily contaminated. The quality of the water available for drinking, agriculture and other uses is vitally important to humans. In this section you will learn about the main sources of water pollution and investigate the procedures used for taking water samples in order to test the levels of contaminants in water supplies and how to treat water to ensure it is safe for human consumption.

### SOURCES OF WATER

Water is found on Earth in three states of matter (gas, liquid and solid), and it readily changes from one state into another. The **water cycle** shown in Figure 15.1.2 involves the continuous movement of water between the land, oceans, streams and atmosphere. Solar energy is the primary energy source for the cycle.



**FIGURE 15.1.2** The water cycle illustrates how water is moved around the Earth through evaporation, condensation and precipitation.

The water cycle involves three main processes.

- Heat from the Sun causes water to evaporate from the oceans, lakes and streams.
- Water vapour in the air is transported around the globe until it condenses to form clouds.
- The water precipitates as rainwater, or occasionally as ice crystals in hail or snow, and falls to the ground.

Human activities such as the combustion of fossil fuels, which produce steam, also contribute to the water cycle.

Table 15.1.1 shows how water is distributed on Earth.

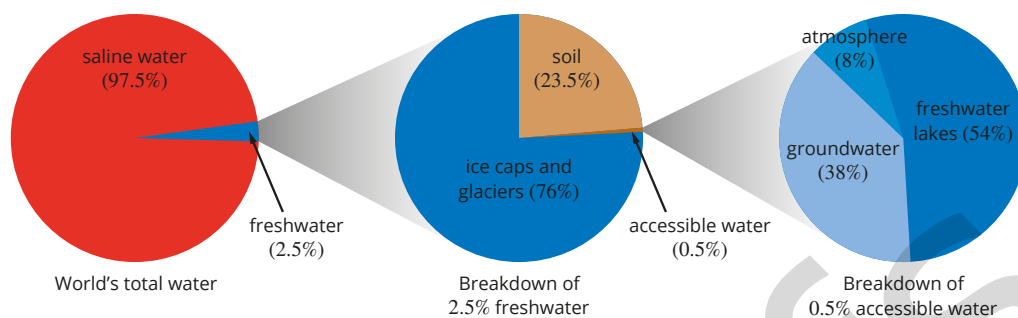
**TABLE 15.1.1** The distribution of water on Earth

| Location of water          | State of matter | Volume (km <sup>3</sup> ) |
|----------------------------|-----------------|---------------------------|
| oceans                     | liquid          | 1 300 000 000             |
| ice caps and glaciers      | solid           | 24 000 000                |
| groundwater                | liquid          | 23 000 000                |
| ground ice and permafrost  | solid           | 300 000                   |
| lakes                      | liquid          | 180 000                   |
| soil moisture              | liquid          | 17 000                    |
| atmosphere as water vapour | gas             | 13 000                    |
| rivers                     | liquid          | 2 100                     |

The volumes in Table 15.1.1 are approximate but, from these figures, the total volume of water on Earth is over 1.3 billion km<sup>3</sup>.

Despite the large quantities of water, there is a limited supply of fresh water on our planet. Figure 15.1.3 shows that only 2.5% of the water on Earth is drinkable and most of this water is not accessible as it is locked in icecaps, glaciers or the soil.





**FIGURE 15.1.3** Distribution of water on Earth. Only 2.5% of the water on Earth is drinkable.

## CHEMFILE

### Great Artesian Basin

The **Great Artesian Basin** is located within Australia. It has the following key features.

- It is the largest **artesian basin** (underground water supply) in the world.
- It covers an area of over 1 700 000 km<sup>2</sup>. It underlies nearly a quarter of the Australian continent.
- In some places, the basin is up to 3000 m deep.
- The temperature of the water in the basin has been recorded to be anywhere from 30°C to 100°C.
- It provides a reliable source of **groundwater** for a very large part of inland Australia.

Traditionally, the water could be readily accessed as it flowed to the surface under natural pressure. However, in the last century, government bodies have set up initiatives to try to maintain stores of water within the basin.



**FIGURE 15.1.4** The Great Artesian Basin provides water for stock and the human population for a large area of inland Australia.

About 40% of the world's population lives in areas where water is scarce and that number will grow to nearly 50% by 2025. Australia has close to 5% of the land mass of the world but has only 1% of the water carried by the world's rivers.

Australia is the driest inhabited continent and has extremely variable rainfall. It has hundreds of large dams and thousands of farm dams that can hold a lot of water so that Australians can be prepared for periods of drought. Since Australia is also very flat and hot, most of the rain that falls evaporates before it can enter rivers and reservoirs.

**i** Due to Australia's harsh climate and low rainfall, Australia has only 1% of the water carried by the world's rivers.

## POTABLE WATER

Obtaining clean drinking water, or **potable water**, has been a challenge for people living in communities for thousands of years.

### Major sources of drinking water

Drinking water in Australia is obtained from a variety of sources. The main sources are:

- reservoirs filled by run-off from rivers and streams
- water obtained directly from rivers and lakes
- groundwater
- **recycled water**
- **desalinated seawater**.

In most of Australia's major cities, water comes from reservoirs built on rivers in protected areas. These reservoirs are able to provide the population with safe, clean water. Water from such sources needs only minor treatment before being released for consumption. Reservoirs and weirs like Mundaring Weir shown in Figure 15.1.5 are surrounded by protected land, classifying them as **protected catchments**. They are often located in national parks and forests with limited access to ensure the quality of the water is very high.

However, in some parts of inland Australia, and in some other countries in the Asia-Pacific region, water comes from sources other than protected catchments. In some cases, water is taken directly from rivers and lakes that may be subject to contamination from run-off from farms and urban areas. Drinking water may also be obtained from groundwater, which is often referred to as **bore water** in Australia. Such sources may contain levels of contamination that require a more complex purification process than for water from protected catchments.





**FIGURE 15.1.5** The Mundaring Weir stores water from the Helena River, near Perth, Western Australia.

Table 15.1.2 shows the major sources of drinking water in some Australian towns and cities.

**TABLE 15.1.2** Sources of drinking water in Australia

| City or town                             | Main water source                              |
|------------------------------------------|------------------------------------------------|
| Alice Springs, inland Northern Territory | groundwater                                    |
| Adelaide, South Australia                | Torrens and Murray rivers                      |
| Bourke, inland New South Wales           | Darling River                                  |
| Broome, coastal Western Australia        | groundwater                                    |
| Melbourne, Victoria                      | reservoirs                                     |
| Mildura, inland Victoria                 | Murray River                                   |
| Birdsville, inland Queensland            | groundwater                                    |
| Hobart, Tasmania                         | Derwent River                                  |
| Perth, Western Australia                 | desalination water, groundwater and reservoirs |

Sources of water in some other countries are shown in Table 15.1.3.

**TABLE 15.1.3** Sources of drinking water in other regions of the world

| Region                      | Main water source                                                                                                     |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Bali, Indonesia             | reservoirs and groundwater                                                                                            |
| Dubai, United Arab Emirates | desalinated seawater (more than 98% sourced this way)                                                                 |
| Hong Kong                   | Dong Jiang River (mainland China), reservoirs (some reclaimed from ocean inlets), and seawater (for flushing toilets) |
| Singapore                   | reservoirs with protected catchments, desalinated seawater, recycled water and water piped from Malaysia              |

**i** Potable water is suitable and safe for human consumption.

## WATER QUALITY

In 2012, Chinese newspapers reported that 54% of China's drinking water was too polluted to drink. Just two years later, the situation in the country had worsened as the percentage of polluted drinking water rose to 60%. About 60 000 people die each year in China from diseases caused by water pollution, such as cholera, dysentery and diarrhoea.

The quality of the drinking water around the world varies enormously. Most of the water in Australia is of high quality but there have been instances where this is not the case.

This section explores the **chemical contaminants** that can be found in drinking water and how samples are obtained to analyse water quality.

### Chemical contaminants

Water is an excellent solvent. It can dissolve many different types of substances including minerals, nutrients, organic material and human-made wastes. This can cause problems for drinking water resources. As rainwater flows down a hillside or stream, or percolates through soil, it will dissolve many different materials on its journey. The dissolved materials may end up in rivers, reservoirs or groundwater, and therefore in Australia's water supplies.

As a result, water supplies may be subject to pollution caused by human activity. Although chemical contamination of water supplies is not common in Australia, it can happen (Figure 15.1.6). Contamination can occur because of:

- run-off from farms and cities
- run-off from industrial and mining wastes
- lead used in solder in copper water pipes.



**FIGURE 15.1.6** Pollution can be caused by seepage of heavy metals and other contaminants into aquifers (water-bearing rock).

Chemical contaminants are elements or compounds that may be harmful if consumed in drinking water. They can be naturally occurring or synthetic. The main types of chemical contaminants are:

- heavy metals
- pollutants from fertilisers
- organic pollutants.

## Heavy metal contamination

**Heavy metals**, such as mercury and lead, can contaminate drinking water and have severe health effects. These effects include cancer, organ and nervous system damage, and, in the most extreme cases, death. Unfortunately, young people are even more susceptible to the effects of heavy metals, and permanent brain damage can be the consequence of exposure to heavy metals. Table 15.1.4 lists some of the common heavy metals and the effects they can have on human health.

**TABLE 15.1.4** Heavy metal contaminants and the risks they pose to human health

| Heavy metal | Source                                                              | Effects                                                                                                                                            |
|-------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| copper      | copper pipes                                                        | anaemia; liver and kidney damage; stomach and intestinal irritation                                                                                |
| lead        | lead pipes, batteries, leaded gasoline (petrol), paints, ammunition | restricts haemoglobin production; damage to kidneys, gastrointestinal tract, joints and reproductive system; can lower IQ levels in young children |
| cadmium     | smelting, improper disposal of rechargeable batteries               | kidney failure; liver disease; osteoporosis                                                                                                        |
| nickel      | power plants, waste incinerators, improper disposal of batteries    | decreased body weight; damage to the heart and liver                                                                                               |
| zinc        | mining, smelting, steel production                                  | anaemia; damage to nervous system and pancreas                                                                                                     |
| arsenic     | mining, smelting, steel production                                  | carcinogenic; stomach pain; numbness; blindness                                                                                                    |
| mercury     | improper disposal of batteries, various industrial processes        | tremors; gingivitis; spontaneous abortion; damage to the brain and central nervous system                                                          |

The quality of drinking water around Australia is governed by local water authorities and is generally high. However, some rural communities do not have access to the same quality of drinking water as city residents do. As a result, there are guidelines in place for the levels of heavy metals that are permissible in Australian drinking water, which are shown in Table 15.1.5.

**TABLE 15.1.5** Australian Drinking Water Guidelines values for heavy metals

| Heavy metal | Guideline value (mgL <sup>-1</sup> ) |
|-------------|--------------------------------------|
| arsenic     | 0.01                                 |
| cadmium     | 0.002                                |
| copper      | 2                                    |
| chromium    | 0.05 as chromium(VI)                 |
| lead        | 0.01                                 |
| mercury     | 0.001                                |



### Example of contamination: mining

In 2013, it was discovered that one third of Tasmania's town water supplies did not meet the national standards for heavy metals. The residents of five towns in the north-west of Tasmania were told not to drink the water due to contamination. In particular, the Tamar River (Figure 15.1.7) was polluted by effluent and the heavy metals zinc, cadmium and lead. The issue for most towns was unusually high concentrations of zinc, cadmium and lead resulting from nearby mining operations.



**FIGURE 15.1.7** The Tamar River in Tasmania has been polluted by effluent, including the heavy metals zinc, cadmium and lead.

### Example of contamination: water pipes

Lead-based solder was once used to join copper water pipes. As a result of corrosion, the lead can be released into water flowing through the pipes. In 1993 in Perth, Western Australia, a study showed that some households had higher than acceptable levels of heavy metals. Table 15.1.6 shows the percentage of houses in Perth with unacceptable heavy metal levels.

To minimise this potential problem, Australian standards were introduced in 1998 to limit the amount of lead in solder to no more than 0.1% by weight.

**TABLE 15.1.6** Heavy metal levels in Perth (1993)

| Heavy metal | Households with unacceptable levels (%) |
|-------------|-----------------------------------------|
| lead        | 5                                       |
| copper      | 12                                      |
| cadmium     | 2                                       |

One method of removing heavy metals from a contaminated water supply is by the use of precipitation reactions. This process involves the following steps.

- 1 The heavy metal cation that is to be removed is identified.
- 2 A soluble compound containing an anion that will form an insoluble compound with the cation of the heavy metal is selected.
- 3 A solution of the compound containing the anions is added to the water.
- 4 The anions combine with the heavy metal cations to form a precipitate.
- 5 The precipitate is filtered off, thereby removing the heavy metal from the water supply.

An example of this process is the addition of calcium hydroxide ( $\text{Ca}(\text{OH})_2$ ) to contaminated water to form a precipitate of a heavy metal hydroxide. Precipitation reactions like this will be looked at in more detail in Chapter 16.

## MONITORING WATER QUALITY

To ensure high water quality and to test for contamination in water supplies, water samples are regularly collected for analysis. The results may be compared against water quality standards. This determines the water's suitability for its intended use and, if necessary, the treatment required to purify the water.

There are protocols for obtaining water samples to ensure that the samples are collected in a consistent manner and with the correct equipment. There are many different methods for obtaining water samples depending upon whether the samples are collected close to shore, from boats (Figure 15.1.8) or from bridges (Figure 15.1.9).



FIGURE 15.1.8 Sampling water from a boat



FIGURE 15.1.9 Sampling water from a bridge

In a large dam or reservoir, it may be necessary to take samples from different depths. When comparing samples over time, samples need to be taken from the same location and in the same manner on each occasion, making it especially important to follow established protocols.

When collecting water samples for testing, it is important to consider the following questions.

- Which chemical is to be analysed and why is the testing to take place?
- What health risks are associated with the sampling?
- What equipment is required, and are **sterile** containers needed?
- What sample size is required for the selected test?
- What method is to be used to record the measurements?
- Where should the samples be taken and at what depths should the samples be taken?
- How will a representative sample be obtained?
- What are the labelling, storage and transport requirements for the sample?

## Protocols for water sampling

The containers used for sampling should not react with the sample. To assure the quality of the sample, the:

- container can be rinsed with the sample before the final sample is taken
- container should be cleaned before taking another sample
- water used for cleaning the container can be tested to ensure there is no contamination between samples.

If the water is to be tested for bacteria and other microorganisms, it is important that the container be sterile.

## Sampling methods and equipment

The method used to avoid contamination of a water sample depends on where the water is being sampled from, and the conditions under which the sample is taken. Some sampling methods are outlined in Table 15.1.7.

**TABLE 15.1.7** Sampling methods for collecting water samples

| Conditions                                                         | Sampling method                                                                                                                                                   |
|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| sample taken from a well-mixed body of water                       | Sampling near the surface is sufficient to obtain a representative sample. A sample should be taken about 10 cm below the surface and away from the water's edge. |
| sample taken from water contaminated by sediment (solid particles) | A sample should be drawn into the sample container by suction to avoid including the sediment.                                                                    |
| sample taken from a river                                          | A sample should be taken upstream from where the person taking the sample stands.                                                                                 |

## Sampling depths

To determine the depth at which a sample should be taken, the temperature is initially measured at every metre of depth.

- If the temperature is consistent, it can be assumed that the water is thoroughly mixed and a sample is taken halfway down.
- If there is a temperature variation, then a sample is taken in the middle of each temperature region.

When the water being sampled is more than 2 metres deep or samples are required from different depths, a Van Dorn sampler (Figure 15.1.10) can be used. This sampler can be lowered horizontally and pulled sideways before the sample is taken. Van Dorn samplers are made of a transparent polymer and are attached by a rope to the surface. After the sampler is raised to the surface, the water is drained and placed in a sterile sample bottle. The Van Dorn sampler is very useful when there are distinct layers in the water, such as in a deep lake or reservoir.

## Sampling locations

To develop a true picture of the water quality in a body of water, you need to take samples from many different locations.

For example, a water-quality monitoring program in the Tamar Estuary in Tasmania took samples from 20 sites divided into five zones, as shown in the map in Figure 15.1.11. The sites ranged from Launceston to where the lower estuary discharges into Bass Strait. The samples were taken monthly at specific depths and all samples were taken on the same day.

## Treatment of drinking water

In most Australian cities and countries in the Asian region, water is treated before being passed on to consumers through the water supply system (Figure 15.1.12). The amount of treatment required varies from city to city, but the usual purpose of the treatment is to remove suspended solids, bacteria, colour and odour from the water.

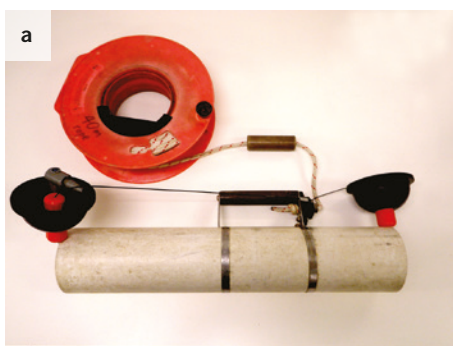
The steps involved in the purification of water for most cities are:

- 1 flocculation
- 2 settling of the floc
- 3 filtering
- 4 chlorination.

The water may also undergo fluoridation. Each of these processes is outlined in more detail.

## Flocculation

**Flocculation** is the process by which small, suspended particles in the water are made to join together to form larger, heavier particles. The heavier particles then sink under their own weight and settle to the bottom of the water sample.



**FIGURE 15.1.10** A Van Dorn sampler consists of a polymer tube with caps at each end.





FIGURE 15.1.11 Sampling sites in the Tamar Estuary, Tasmania

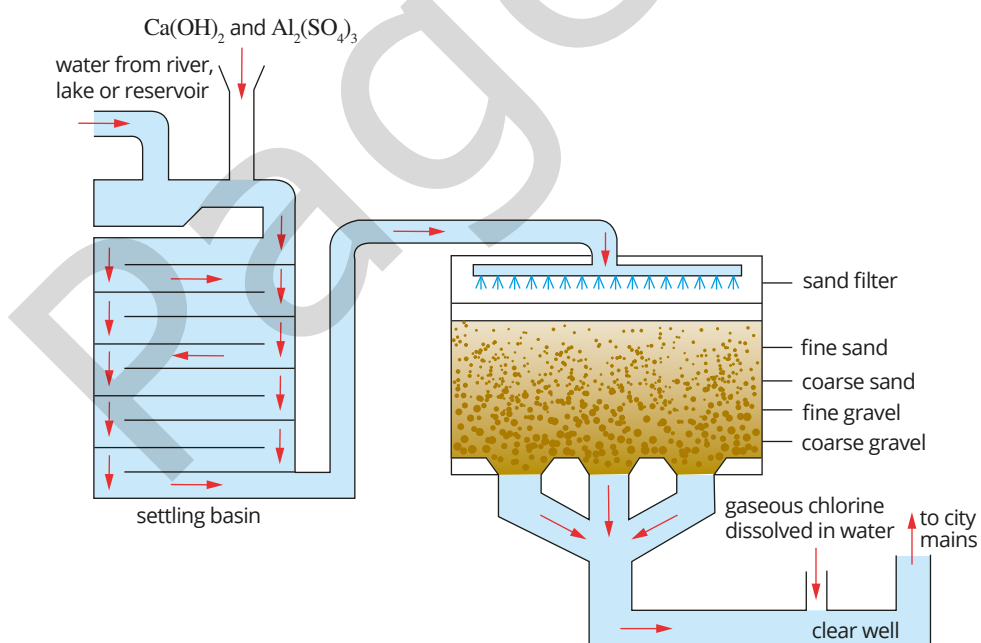
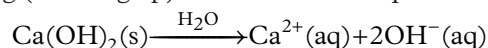
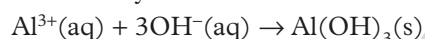


FIGURE 15.1.12 A typical system for treating drinking water

Flocculation can be achieved by adding alum (aluminium sulfate) and, if necessary, lime. Lime ( $\text{Ca}(\text{OH})_2$ ) is added to neutralise acids and provide a source of hydroxide ions in the water supply. Lime is an ionic compound that dissolves in water by dissociating (breaking up) into its ions. The equation for this dissociation is:



Alum provides  $\text{Al}^{3+}$  ions in solution and these combine with hydroxide ions to form a precipitate of aluminium hydroxide:



Aluminium hydroxide is produced in the form of a gelatinous (jelly-like) precipitate called the **floc**, which traps other fine particles and removes colour and some microorganisms from the water. These other materials are adsorbed in the aluminium hydroxide precipitate. The finely divided precipitate particles then coagulate (clump together) to form heavier particles.

### Settling, or sedimentation

**Settling**, or sedimentation, utilises gravity to remove suspended solids from the water. After flocculation, the water is left to stand to allow the floc to settle for a period, during which the settled materials form a sludge. The sludge accumulates at the bottom of the settling tank and is removed. The remaining water passes onto the filtering stage.

### Filtration

Following settling, water from the settling tank is allowed to filter through a bed of sand over gravel. This removes any remaining suspended matter.

### Chlorination

After filtering, the clear water is usually treated with gaseous chlorine to destroy any bacteria. The main purpose of **chlorination** is to remove biological contaminants. After chlorination, water is considered fit for human consumption.

### Fluoridation

In all the capital cities and the majority of other cities in Australia, fluoride is added to drinking water in a process called **fluoridation** before it is released from storages. Fluoride in drinking water helps to reduce the incidence of tooth decay. It does this by interacting with tooth enamel ( $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$ ). The fluoride ion replaces the hydroxyl ion in tooth enamel, forming fluorapatite ( $\text{Ca}_{10}(\text{PO}_4)_6\text{F}_2$ ) which is stronger and more resistant to decay.

Fluoride is added to the water in the form of compounds including sodium hexafluorosilicate ( $\text{Na}_2\text{SiF}_6$ ) fluorosilicic acid ( $\text{H}_2\text{SiF}_6$ ) or sodium fluoride ( $\text{NaF}$ ). Each of these compounds releases fluoride ions when it dissolves in water.

The water obtained from these procedures is suitable for drinking and is described as potable.

## CHEMISTRY IN ACTION

# Obtaining drinking water from seawater

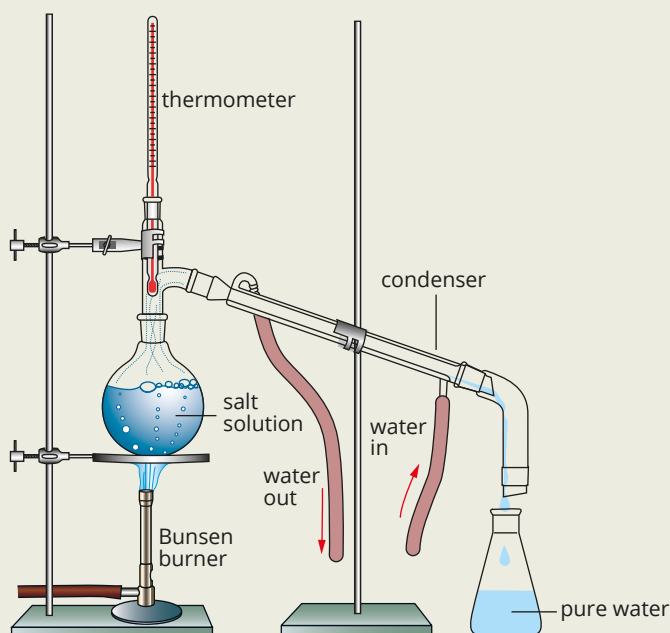
Fresh water is a relatively scarce commodity on Earth, whereas seawater is plentiful. Seawater contains about 3.5% dissolved salts and so is not suitable for either drinking or agriculture. **Desalination** involves the removal of dissolved salts from seawater to obtain fresh water. Two main desalination processes are:

- distillation
- reverse osmosis.

## Producing drinking water by distillation

Figure 15.1.13 illustrates simple laboratory distillation equipment that can be used to obtain fresh water from salt water.

- Salt water is placed in the round-bottomed flask on the left of the diagram, and then heated until it boils. Water vapour passes out from the flask and over into the condenser. Salt is left behind in the flask and so the salt and water vapour are separated.



**FIGURE 15.1.13** Distillation can be used to obtain fresh water from salt water.

- The condenser is a glass tube with a water-cooled 'jacket' surrounding it. As the water vapour passes down the condenser tube it is cooled and condenses back into water.
- The now pure water is collected in the conical flask at the end of the condenser.

Distillation is an expensive way to produce fresh water because of the cost of fuel to provide the heat required. In Australia, this is not a viable way to produce drinking water on a large scale.

The situation is different, however, in the United Arab Emirates, a country that has large supplies of natural gas. Dubai is a coastal city in the United Arab Emirates. The surrounding land is very dry and water is scarce. The city of Dubai has therefore turned to the desalination of seawater as a source of drinking water. In the Jebel Ali desalination plant near Dubai shown in Figure 15.1.14,



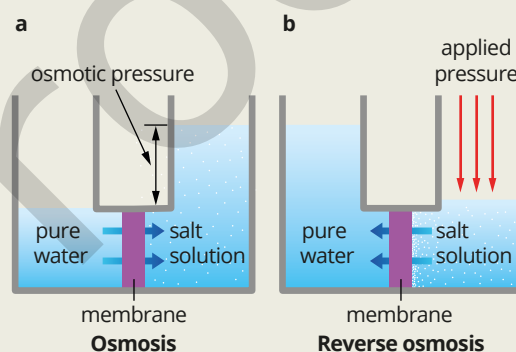
**FIGURE 15.1.14** The Jebel Ali desalination plant has six gas turbines that generate power from burning natural gas.

gas turbines generate electricity and the waste heat is used to produce fresh water from seawater in a distillation process. Over 98% of Dubai's water supply is provided by desalination.

### Producing drinking water by reverse osmosis

**Osmosis** is the natural tendency of water to move from a region of low salt concentration to one of higher salt concentration. If salt water and fresh water are separated by a semipermeable membrane that allows water molecules, but not the dissolved ions, to pass through it, the water molecules will pass through the membrane from the fresh water to the salt water.

This natural tendency can be reversed if pressure is applied to the saltwater side of the semipermeable membrane. This process is called **reverse osmosis** and can be seen in Figure 15.1.15.



**FIGURE 15.1.15** (a) Natural osmosis; (b) Under pressure, reverse osmosis can be used to obtain fresh water from seawater.

The difficulty and associated cost of reverse osmosis is in making suitable membranes that will not rupture under the high pressures used.

A number of reverse osmosis plants were built in Australia during the extended drought of the first decade of this century. They can be found in Western Australia, South Australia, Victoria, New South Wales and Queensland (Figure 15.1.16).



**FIGURE 15.1.16** Desalination plant in Kwinana, Western Australia



## 15.1 Review

### SUMMARY

- Most of the water on Earth is in liquid form and contained in the oceans.
- Only 2.5% of all the water on Earth is drinkable, but just 0.5% of all the water on Earth is drinkable and accessible.
- Drinking water can be obtained from a variety of sources including:
  - rivers flowing through protected catchments
  - directly from rivers and lakes
  - groundwater
  - rainwater collected from a roof and stored in tanks
  - desalinated seawater.
- Reservoirs, which are fed by rivers, are the main source of household water in Australian cities.
- Contamination of water supplies comes from farms, industry, mines and through the soil.
- Types of chemical contaminants include:
  - heavy metals
  - pollutants from fertiliser
  - organic pollutants.
- Protocols exist for sampling water for the analysis of water quality.
- All protocols need to consider:
  - the reason for testing
  - the chemical to be analysed
  - health risks associated with sampling
  - specific requirements for the chemical being tested
  - selection of equipment and sample size
  - recording results and taking representative samples
  - labelling and storage requirements.
- Drinking water from reservoirs is usually purified using a sequence of steps including flocculation, settling, filtration and chlorination and fluoridation.
- Drinking water is often fluoridated to reduce dental decay.
- The two main methods of desalination of seawater are:
  - distillation
  - reverse osmosis.
- Elevated levels of heavy metals in drinking water can cause problems for human health.
- Heavy metals may be removed from waste water by chemical processes including precipitation.

### KEY QUESTIONS

- 1 Which water source has the highest risk of contamination? Explain why.
- 2 Why is only 0.5% of water on Earth available for drinking?
- 3 Why does Australia store more water per person than any other country?
- 4 In a large body of water, the temperature of water should be measured at every metre of depth before taking samples. What is the reason for doing this?
- 5 Drinking water can be produced by distillation of salt water. Which of the following statements is correct? (There may be more than one.)
  - A Salt water is placed in a flask and heated until it boils.
  - B Water vapour passes out of the flask and over into a condenser.
  - C The condenser is a tube with a water-cooled jacket surrounding it.
  - D Pure water is collected into the flask that originally held the salt water.
- 6 Explain why fluoridation is an accepted additional treatment of drinking water in many Australian cities.

## 15.2 Properties of water

Water and life are so strongly linked that space scientists search the universe for water in their quest to discover possible life beyond our planet. It is the unique properties of water that allow it to support life. Some of the ways water supports life are summarised in Table 15.2.1.

**TABLE 15.2.1** The properties of water allow it to sustain life.

| Role water plays in sustaining life                                                                         | Property of water that allows it to perform this role                                                                        |
|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| provides an internal transport system in plants and animals                                                 | ability to dissolve many substances easily                                                                                   |
| provides a system to transport heat from cells to the body's surface of animals                             | high specific heat capacity                                                                                                  |
| provides a cooling system in animals and plants by evaporation of water from skin and the surface of leaves | high latent heat of vaporisation                                                                                             |
| provides an environment for aquatic plants and animals                                                      | expands on freezing, so provides an insulating surface layer on ponds and rivers; ability to dissolve many substances easily |
| reactant for photosynthesis                                                                                 | ability to react with carbon dioxide to form glucose and oxygen                                                              |

The unique properties of water can be explained by looking at the structure of a water molecule and the hydrogen bonding that occurs between the water molecules.

### WATER AND HYDROGEN BONDING

#### Structure of water

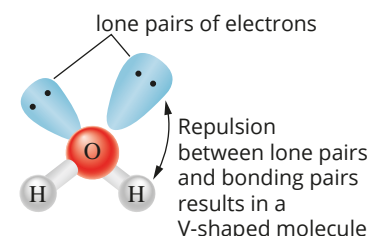
Water has the chemical formula  $\text{H}_2\text{O}$ , which means that each water molecule contains two hydrogen atoms and one oxygen atom. Hydrogen and oxygen are both non-metals so they bond with each other by sharing electrons. Each hydrogen atom shares one electron with the oxygen atom and the oxygen atom shares one electron with each hydrogen atom. The sharing of electrons means these intramolecular bonds are covalent.

Each oxygen atom also has two lone pairs (non-bonding pairs) of electrons that contribute to the molecule's shape. A water molecule is V-shaped (or bent) as shown in Figure 15.2.1. The lone pairs of electrons on the oxygen atom have greater repulsion than the electrons forming the single bonds between oxygen and hydrogen. This pushes the hydrogen atoms closer together, giving the overall bent shape (the shape of molecules was explained in detail in Chapter 12).

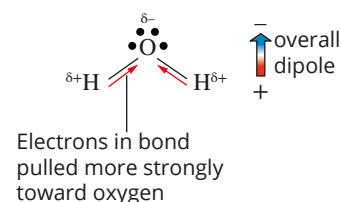
The covalent bonds in a water molecule are polar. The oxygen atom has a higher electronegativity than the hydrogen atoms. This means that the electrons in the O–H bonds are more strongly attracted to the oxygen atom. Uneven sharing of the electrons gives the oxygen atom a partial negative ( $\delta^-$ ) charge and each hydrogen atom carries a partial positive charge ( $\delta^+$ ).

Since the bonding electrons are distributed unevenly across the water molecule, the molecule can be described as a dipole. Figure 15.2.2 shows how the individual dipoles of the O–H bonds in water add together, resulting in an overall molecular dipole.

Polar bonds, polar molecules and dipoles were explained in detail in Chapter 12.

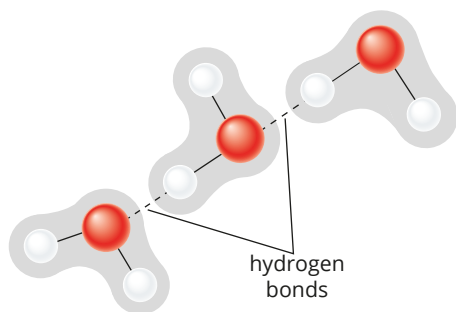


**FIGURE 15.2.1** In this diagram of the structure of water, the two hydrogen atoms (white) form a V-shape with a central oxygen atom (red).



**FIGURE 15.2.2** A water molecule has an overall dipole. The oxygen atom has a partial negative charge while the hydrogen atoms both have partial positive charges.

**i** Water molecules are polar and have a permanent overall dipole.



**FIGURE 15.2.3** The hydrogen bonding between molecules of water.

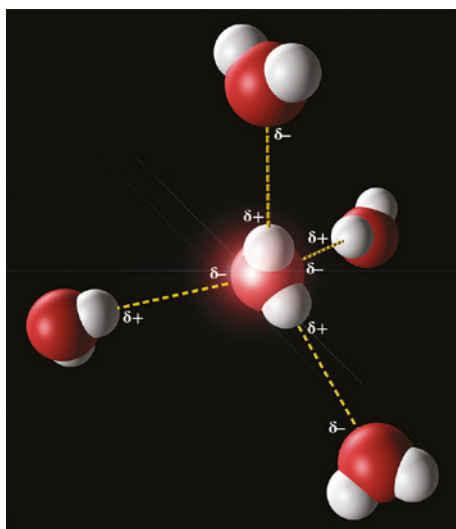
## Hydrogen bonding

The main type of intermolecular force between molecules in water is a hydrogen bond. The hydrogen bonds are formed by an electrostatic attraction between the partial positive charge on a hydrogen atom on one water molecule and a lone pair of electrons on the oxygen atom of a neighbouring water molecule.

Figure 15.2.3 shows the hydrogen bonding between molecules of water. The two hydrogen atoms in each water molecule have a slight positive charge, while the central oxygen atom has a slight negative charge. The black dashed lines represent the hydrogen bonds that form between water molecules.

Because there is a large electronegativity difference between oxygen and hydrogen atoms, the partial charges on a water molecule are relatively large. The electrostatic attraction between these opposite partial charges makes the hydrogen bonds between water molecules quite strong.

As Figure 15.2.4 shows, each water molecule can form hydrogen bonds with up to four other water molecules.



**FIGURE 15.2.4** Each water molecule can form hydrogen bonds to four other water molecules.

## WATER HAS RELATIVELY HIGH MELTING AND BOILING POINTS

Compared to other molecules of a similar size, water has high boiling and melting points. This can most easily be seen through the observed trends in these properties for the group 16 hydrides.

### Trends in group 16 hydrides

The group 16 elements include oxygen (O), sulfur (S), selenium (Se), tellurium (Te) and polonium (Po). Each of these elements can bond with hydrogen to form compounds known as **hydrides**. Water is a group 16 hydride. Others are listed in Table 15.2.2.

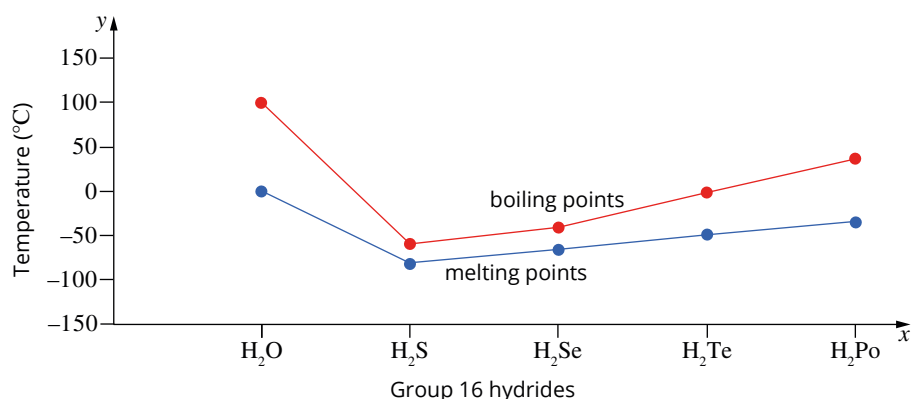
**TABLE 15.2.2** Names and formulae of the group 16 hydrides

| Element | Name and formula of hydride | Formula           |
|---------|-----------------------------|-------------------|
| O       | water                       | H <sub>2</sub> O  |
| S       | hydrogen sulfide            | H <sub>2</sub> S  |
| Se      | hydrogen selenide           | H <sub>2</sub> Se |
| Te      | hydrogen telluride          | H <sub>2</sub> Te |
| Po      | hydrogen polonide           | H <sub>2</sub> Po |

The group 16 hydrides are all molecular compounds, so their melting and boiling points reflect the size of the forces between their molecules. The higher the melting and boiling points, the stronger the intermolecular forces must be because more energy is required to overcome the forces and allow the molecules to move apart from each other.

The melting and boiling points of the group 16 hydrides are shown in Table 15.2.3 and Figure 15.2.5.





**FIGURE 15.2.5** Graph showing the trend in melting and boiling points of water and other group 16 hydrides

You can see from Figure 15.2.5 that, apart from water, there are clear trends in the melting and boiling points of the group 16 hydrides. The melting and boiling points both increase going down the group. This increase indicates that the intermolecular forces are also getting stronger down the group. The intermolecular forces responsible for this trend are dispersion forces. Dispersion forces increase in strength with increasing mass. H<sub>2</sub>Po has a larger mass than H<sub>2</sub>S and H<sub>2</sub>Se; hence it has stronger dispersion forces and higher melting and boiling points.

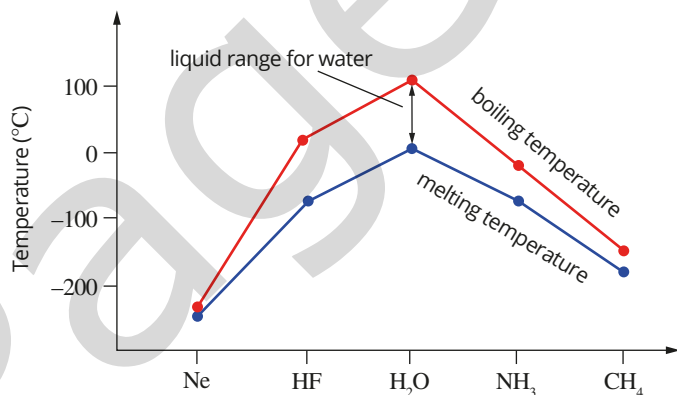
**i** Despite being the smallest hydride, water has significantly higher melting and boiling points than other group 16 hydrides.

**TABLE 15.2.3** Melting and boiling points of the group 16 hydrides

| Hydride           | Melting point (°C) | Boiling point (°C) |
|-------------------|--------------------|--------------------|
| H <sub>2</sub> O  | 0                  | 100                |
| H <sub>2</sub> S  | -82                | -60.7              |
| H <sub>2</sub> Se | -66                | -41.5              |
| H <sub>2</sub> Te | -49                | -2.2               |
| H <sub>2</sub> Po | -35                | 36.1               |

## The melting and boiling points of water are exceptional

Water has a melting point of 0°C and a boiling point of 100°C. Both of these values are significantly higher than those of other group 16 hydrides. The values are also significantly higher than those of other molecular substances of a similar size, as can be seen in Figure 15.2.6.



**FIGURE 15.2.6** The melting and boiling points of water and other molecules of similar size

It is the relatively strong hydrogen bonds between its molecules that give water its exceptional properties.

**i** Each water molecule has the potential to form four hydrogen bonds with surrounding water molecules. There are two partially charged hydrogen atoms and two lone pairs of electrons on the oxygen atom in each molecule so all the hydrogen atoms and lone pairs in the molecule can be involved in hydrogen bonding.

A significant amount of energy is needed to disrupt all of the hydrogen bonds between water molecules, resulting in the higher melting points and boiling points observed for water.

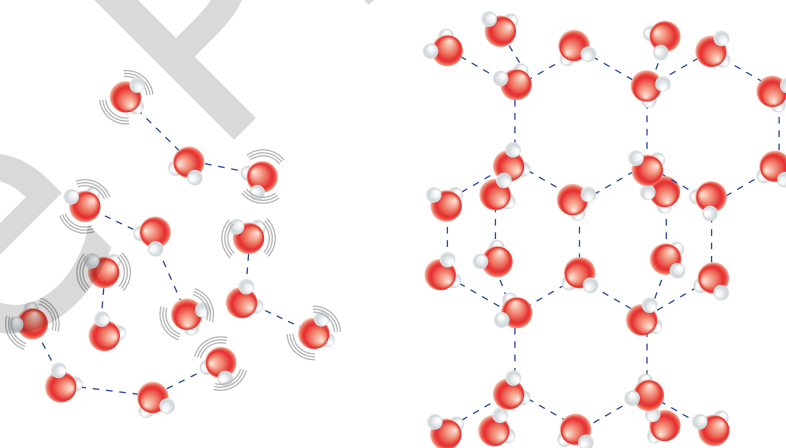
## DENSITY IN THE LIQUID AND SOLID STATES

### Water expands on freezing

When a sample of water freezes, the resulting ice crystal lattice has a greater volume than when the sample of water was a liquid. Water has the unusual property of having a lower density in the solid phase than in the liquid phase.

This property of water is important to life on Earth. When water at the surface of rivers or lakes freezes, the low density of the ice means it remains on the surface. This surface layer of ice insulates the water below from the cold air temperatures, reducing the possibility that the entire river or lake will freeze, and allowing aquatic life to survive.

As liquid water is cooled, the water molecules move more slowly. Upon approaching the freezing temperature of water, the molecules arrange in a way where each water molecule forms four hydrogen bonds to four neighbouring water molecules, as shown in Figure 15.2.7.



**FIGURE 15.2.7** The arrangement of water molecules in liquid water (left) and ice (right)

This creates a very open arrangement of molecules, meaning that the water molecules in ice are more widely spaced than in liquid water. Therefore, ice is less dense than liquid water. When ice melts, the water molecules move more freely and move closer together.

### High surface tension of water

The water molecules at the surface of a sample of water are not completely surrounded by other water molecules. As a result, they form hydrogen bonds with neighbouring molecules on the sides and below, but not above.

The sideways forces of attraction to neighbouring molecules are equal in all directions, so for molecules in the body of the liquid, there is no net force. However, the attraction of surface water molecules to the molecules below them has no opposite force, so the water molecules at the surface are pulled downwards.

In a beaker of water at room temperature, a water molecule will spend, on average, only  $10^{-9}$  seconds or 1 nanosecond at the surface before being pulled back into the bulk of the liquid (Figure 15.2.8). The surface of the water is in a constant state of tension.

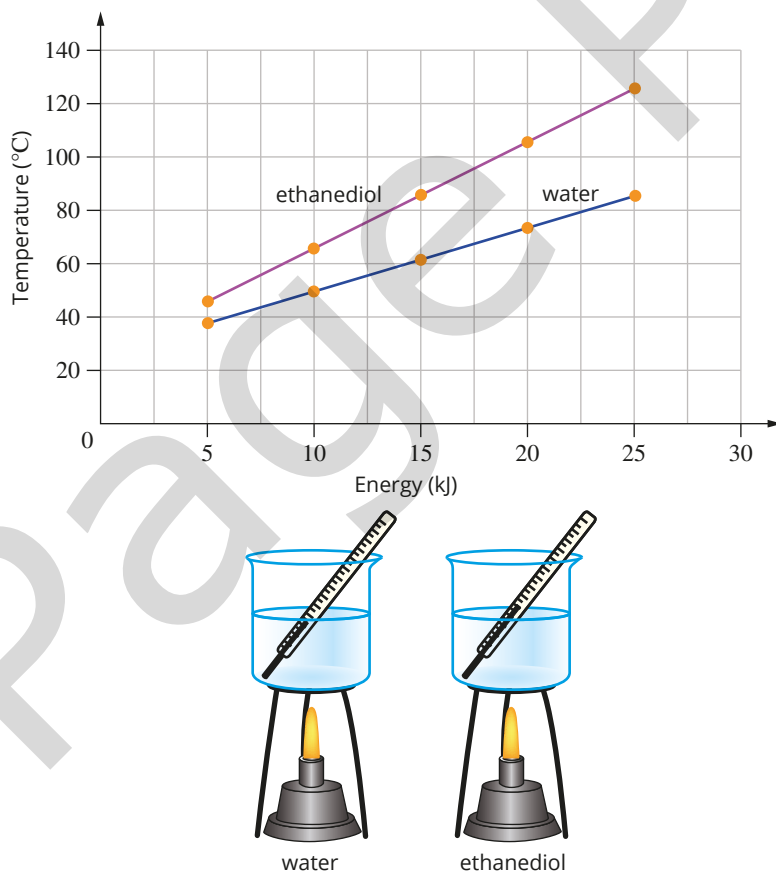
Water, with relatively strong hydrogen bonds between molecules, has a relatively high surface tension. The surface tension of a substance is a measure of the resistance of a liquid to increasing its surface area. The surface tension is why falling water tends to form droplets. It also explains why insects are able to walk across water.

In Figure 15.2.9 the surface of the water behaves like a stretched skin. The water molecules at the surface are so attracted to each other by strong hydrogen bonds that the paperclip floats, even though the density of the paperclip is higher than the density of water.

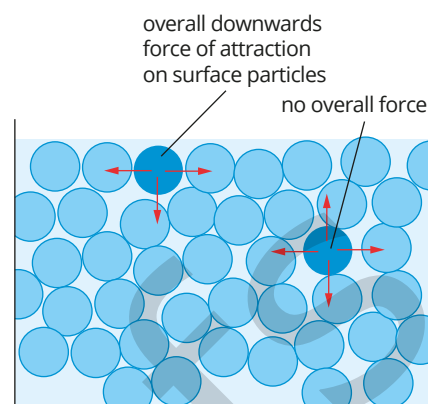
## HEAT CAPACITY

The **heat capacity** of a substance is a measure of the substance's capacity to absorb and store heat energy. It is a measure of how much energy the substance absorbs as its temperature increases. When the same quantity of heat energy is applied to two substances with different heat capacities, they will undergo different temperature changes.

This can be seen with water and ethanediol. Water has a higher specific heat capacity than ethanediol. When you heat equal amounts of the two liquids with the same amount of heat energy, the temperature of the two liquids will rise by different amounts. As shown in Figure 15.2.11, when 100 g of water and 100 g of ethanediol are heated with the same amount of energy, the beaker containing water will increase in temperature by a smaller amount than the ethanediol.



**FIGURE 15.2.11** When 100 g of water and 100 g of ethanediol are heated with the same amount of energy, the temperature of the beaker containing water will increase less than the temperature of the ethanediol.



**FIGURE 15.2.8** The forces of attraction on particles at the surface and in the body of a liquid. The particles in the body of the liquid experience no net force. The particles at the surface of the liquid have an overall downwards force of attraction.



**FIGURE 15.2.9** The relatively high surface tension of water allows a paperclip to float.

## CHEMFILE

### The deep blue sea

When light passes through water, the vibration of O–H bonds in water molecules, combined with the hydrogen bonds between molecules, causes a small amount of light to be absorbed in the red part of the visible colour spectrum. This leaves the remaining complementary colour, which is blue, for you to see (Figure 15.2.10). You cannot see this colour in small volumes of water. However, when light travels through several metres of water or ice, you can see the blue colour.



**FIGURE 15.2.10** Large volumes of water or ice appear blue.

## Specific heat capacity

The **specific heat capacity** of a substance measures the amount of energy (in joules) needed to increase the temperature of a certain amount (usually 1 gram) of that substance by 1°C.

Specific heat capacity is given the symbol  $C$  and is expressed in joules per grams per degrees Celsius, i.e.  $\text{J g}^{-1} \text{ } ^\circ\text{C}^{-1}$ .

**i** Specific heat capacity ( $C$ ) is a measure of the energy required to raise 1 g of substance by 1°C. It is reported in units of  $\text{J g}^{-1} \text{ } ^\circ\text{C}^{-1}$ .

The specific heat capacities of some common substances are listed in Table 15.2.4.

**TABLE 15.2.4** Specific heat capacities of common substances

| Substance  | Specific heat capacity ( $\text{J g}^{-1} \text{ } ^\circ\text{C}^{-1}$ ) |
|------------|---------------------------------------------------------------------------|
| water      | 4.18                                                                      |
| ethanediol | 2.42                                                                      |
| ethanol    | 2.40                                                                      |
| sand       | 0.48                                                                      |
| copper     | 0.39                                                                      |
| lead       | 0.16                                                                      |

The specific heat capacity of a substance is a reflection of the types of bonds holding the molecules, ions or atoms together in the substance. The relatively high specific heat capacity of water is related to the number and strength of the hydrogen bonds between water molecules.

**i** The specific heat capacity of a substance reflects the type of bonding in that substance. For covalent molecules, this will depend on the strength of the intermolecular forces between molecules.

## Specific heat capacity of water

Water has a specific heat capacity of  $4.18 \text{ J g}^{-1} \text{ } ^\circ\text{C}^{-1}$ . This means that 4.18 joules of heat energy are needed to increase the temperature of 1 gram of water by 1°C.

The specific heat capacity of water is relatively high. This is because of the presence of hydrogen bonds between water molecules. Hydrogen bonds are stronger than other intermolecular forces and they are able to absorb and store large amounts of heat energy before they break.

As water is a liquid, it is often convenient to measure a quantity of water as a volume, in mL or L, rather than as a mass measured in grams. 1 mL of water has a mass of 1 g, so it is easy to convert a given volume of water to mass:

- 1 mL of water weighs 1 g.
- 100 mL of water weighs 100 g.
- 250 mL of water weighs 250 g.

**i** The density of liquid water is  $1 \text{ g mL}^{-1}$ , so volumes of water in millilitres (mL) are equal to the mass of water in grams (g).



**EXTENSION**

## Calculations using specific heat capacity

The specific heat capacity of a substance can be used to calculate the heat energy in joules required to increase the temperature of a given mass of substance by a particular amount. Heat energy is given the symbol  $q$ .

A useful equation can be written:

$$\text{Heat energy} = \text{specific heat capacity} \times \text{mass} \times \text{temperature change}$$

Using symbols, the equation can be written:

$$q = C \times m \times \Delta T$$

where  $q$  is the amount of heat energy (J),  $C$  is the specific heat capacity ( $\text{J g}^{-1} \text{ } ^\circ\text{C}^{-1}$ ),  $m$  is the mass (g) and  $\Delta T$  is the temperature change ( $^\circ\text{C}$ ).

### Worked example 15.2.1

#### CALCULATING THE AMOUNT OF ENERGY REQUIRED TO HEAT A SPECIFIED MASS OF A SUBSTANCE USING SPECIFIC HEAT CAPACITY

Calculate the heat energy, in kJ, needed to increase the temperature of 200 g of water by  $15.0^\circ\text{C}$ .

| Thinking                                                                                                 | Working                                                                                       |
|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Find the specific heat capacity ( $C$ ) of the substance from Table 12.2.1.                              | The specific heat capacity of water is $4.18 \text{ J g}^{-1} \text{ } ^\circ\text{C}^{-1}$ . |
| To calculate the quantity of heat energy in joules, use the formula:<br>$q = C \times m \times \Delta T$ | $q = 4.18 \times 200 \times 15.0$<br>$= 1.25 \times 10^4 \text{ J}$                           |
| Express the quantity of energy in kJ. Remember that to convert from J to kJ, you multiply by $10^{-3}$ . | $q = 1.25 \times 10^4 \times 10^{-3}$<br>$= 12.5 \text{ kJ}$                                  |

### Worked example: Try yourself 15.2.1

#### CALCULATING THE AMOUNT OF ENERGY REQUIRED TO HEAT A SPECIFIED MASS OF A SUBSTANCE USING SPECIFIC HEAT CAPACITY

Calculate the heat energy, in kJ, needed to increase the temperature of 375 g of water by  $45.0^\circ\text{C}$ .

**CHEMFILE**

### Biological importance of water's high heat capacity

The high heat capacity of water means water will absorb a relatively large amount of energy before it heats up. This makes water resistant to sudden temperature changes, and plants and animals that live in water survive without experiencing wide temperature fluctuations.

Many living organisms are mainly composed of water. Water's high heat capacity allows organisms to maintain consistent internal body temperatures, regardless of the external environment.



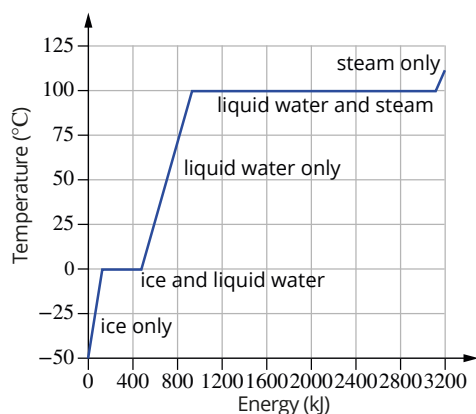
**FIGURE 15.2.12** Fish and coral in the sea can survive due to the relatively consistent temperature of the sea water. Water's high heat capacity makes it resistant to temperature fluctuations.

## LATENT HEAT

When a solid is heated at constant pressure, the temperature of the solid increases until it reaches its melting point. The temperature then remains constant as the solid melts, even though further energy is being absorbed.

Similarly, when a liquid reaches its boiling point, its temperature remains constant until all of the liquid has evaporated. Think about a pot of boiling water. The temperature remains at  $100^\circ\text{C}$  until all of the water in the pot has evaporated.

Figure 15.2.13 shows how in both cases, at the melting point and boiling point of the substance, the temperature remains constant until the phase change is complete. Heat is still flowing into the substance but the temperature does not change. Once a solid has completely melted, or a liquid has completely evaporated, if heating continues, then the temperature of the substance starts to increase again.



**FIGURE 15.2.13** Latent heat is the energy absorbed by a substance in order to change state at its melting or boiling temperature. For a pure substance, such as water in this case, the horizontal regions of the heating curve show where latent heat is being absorbed and the temperature is not changing.

Even though the temperature does not change while the phase change is taking place, the substance is still absorbing energy. This energy is referred to as **latent heat** and is defined as the energy absorbed by a fixed amount of substance as it changes state from a solid to a liquid or a liquid to a gas at its melting point or boiling point respectively.

**i** Latent heat is the energy required to change a fixed amount of substance, usually 1 mole, from either a solid to a liquid or a liquid to a gas. Over the period of time that latent heat is being absorbed, the temperature of a substance will not change.

## Values of latent heat

Latent heat values are a measure of the quantity of heat energy required to melt or boil a given amount of a solid or liquid. Latent heat values have the unit kilojoules per mole,  $\text{kJ mol}^{-1}$ .

The **latent heat of fusion** of a substance is the heat needed to change 1 mole of the substance from a solid to a liquid at its melting point.

The **latent heat of vaporisation** of a substance is the heat needed to change 1 mole of the substance from a liquid to a gas at its boiling point.

## Latent heat of water

The latent heat of fusion of water is  $6.0 \text{ kJ mol}^{-1}$ . This means that  $6.0 \text{ kJ}$  of energy is needed to change 1 mole of water from a solid to a liquid at  $0^\circ\text{C}$ . This energy is needed to disrupt the ice lattice by breaking some of the hydrogen bonds between water molecules.

The latent heat of vaporisation of water is  $44.0 \text{ kJ mol}^{-1}$ . This means that  $44.0 \text{ kJ}$  is needed to change the state of 1 mole of water from a liquid to a gas at  $100^\circ\text{C}$ . This relatively large quantity of energy is required to completely break the hydrogen bonds between the water molecules so they can separate and form a gas.

The latent heat values of water and some other substances are listed in Table 15.2.5. Much like specific heat capacity, latent heat depends on the strength of the intermolecular forces between molecules of the substance.

**TABLE 15.2.5** Latent heat values for some common molecular substances

| Substance | Latent heat of fusion ( $\text{kJ mol}^{-1}$ ) | Latent heat of vaporisation ( $\text{kJ mol}^{-1}$ ) |
|-----------|------------------------------------------------|------------------------------------------------------|
| water     | 6.0                                            | 44.0                                                 |
| hydrogen  | 0.06                                           | 0.45                                                 |
| oxygen    | 0.22                                           | 3.4                                                  |

As you can see, the latent heat values of water are relatively high. This is due to the strength and number of water's hydrogen bonds relative to its molecular size.

## 15.2 Review

### SUMMARY

- A water molecule is V-shaped and contains polar covalent bonds.
- Water molecules are polar because an overall dipole exists as a result of the asymmetrical nature of the polar O–H bonds.
- As a result of the O–H bonds in water molecules, the forces that attract one water molecule to another are relatively strong hydrogen bonds.
- Each water molecule can form up to four hydrogen bonds with other water molecules.
- The strength of water's hydrogen bonds means relatively large amounts of energy are required to disrupt the bonds and separate the molecules from each other. This gives water relatively high melting and boiling points.
- Water has significantly higher melting and boiling points than the other group 16 hydrides.
- With the exception of water, the melting and boiling points of the group 16 hydrides increase going down the group. This is due to the increasing strength of dispersion forces.
- Ice is less dense than liquid water because of its unique geometric arrangement of water molecules as a result of hydrogen bonding.
- Heat capacity is a measure of a substance's capacity to absorb and store heat energy.
- The specific heat capacity of a substance measures the quantity of energy (in joules) needed to increase the temperature of a certain amount (usually 1 gram) of that substance by 1°C.
- The specific heat capacity of water is relatively high due to the ability of the hydrogen bonds between water molecules to absorb and store heat energy.
- The mass of a sample of liquid water in grams (g) is equal to its volume in millilitres (mL).
- Latent heat is the energy absorbed by a fixed amount of substance as it changes state from a solid to a liquid or a liquid to a gas.
- The latent heat of vaporisation of water is relatively high. This has significance for organisms and water supplies.

### KEY QUESTIONS

- a List the physical properties that make water unique.
  - b Explain the significance of polarity and hydrogen bonding in relation to these properties of water.
- 2 Explain why a water molecule can form up to four hydrogen bonds with other water molecules.
- a Describe the forces that must be overcome in order for ice to melt.
  - b Sketch, or describe, a portion of a lattice of ice.
  - c Explain why ice is less dense than liquid water.
- 4 Which one of the following statements best describes the specific heat capacity of a substance?
  - A Specific heat capacity is a measure of the heat energy required to increase the temperature of a substance from its melting point to its boiling point.
  - B A substance with high specific heat capacity is less resistant to temperature change than a substance with low heat capacity.
  - C A substance with high specific heat capacity contains bonds that are unable to absorb a lot of energy.
  - D Specific heat capacity is a measure of the heat energy that can be absorbed by the bonds in a substance.
- 5 Select the overall force experienced by a water molecule at the surface of a glass of water.
  - A up
  - B down
  - C sideways
  - D no force at all
- 6 A heating curve for a substance is produced by plotting the temperature change against the energy input. When this is done from the melting point to the boiling point of a substance, two flat regions are observed. Explain what the flat regions of the graph represent.
- 7 What type of bonding is responsible for water's relatively high latent heat values?

## 15.3 Water as a solvent



**FIGURE 15.3.1** When solid potassium permanganate is added to water, it dissolves. The particles disperse into the solution and move around freely.

Water is an excellent solvent. This is one of its most important properties. Almost all biological processes and many industrial processes occur in water. These systems are known as **aqueous** environments.

When substances are dissolved in water, the particles are free to move throughout the **solution**. When two aqueous reactants are combined in a reaction vessel, the dissolved reactant particles mix freely. This increases the chances of the reactants coming into contact. Because of the increased movement of the reactant particles, interactions between reactants are generally much more effective than if the same reactants were mixed as solids.

In Figure 15.3.1, you can see how the deep purple colour of potassium permanganate spreads through the water as the solid dissolves. Eventually the liquid will appear completely purple as the particles continue to mix and move.

This section looks at the importance of the solvent properties of water to living organisms and modern industry, as well as the process of dissolving. Sections 15.4 and 15.5 will explore in more detail the type of bonds that form when molecular and ionic substances dissolve in water.

### FORCES BETWEEN SOLUTE AND SOLVENT PARTICLES

#### Aqueous solutions

An aqueous solution is formed when a solid, liquid or gas is dissolved in water. The substance being dissolved is called the **solute** and the liquid in which the substance is dissolved is the **solvent**.

The solute of an aqueous solution can change but the solvent will always be water. You can have an aqueous solution of salt (saline) or an aqueous solution of sugar. Table 15.3.1 lists some common aqueous solutions.

**TABLE 15.3.1** Some everyday aqueous solutions

| Solution                                      | Solute/s                               | Solvent |
|-----------------------------------------------|----------------------------------------|---------|
| saline solution (for use with contact lenses) | sodium chloride                        | water   |
| soft drink                                    | carbon dioxide, sugar, flavour, colour | water   |
| coffee                                        | coffee, sugar                          | water   |

All solutions have the following characteristics.

- The solute and the solvent cannot be distinguished from each other. This means the solution is **homogeneous**.
- The dissolved particles are too small to see.
- The amount of dissolved solute can vary from one solution to another.

**i** In a homogeneous solution, the solute and solvent particles cannot be distinguished from each other. Every part of the solution is the same as any other part.



## EXTENSION

# Suspensions and colloids

When a substance is mixed with water, it does not always dissolve to form a solution. It could also form a suspension or a colloid.

A suspension is a heterogeneous mixture (a mixture with visible distinguishable parts) that forms when a solute does not dissolve significantly in a solvent. Some particles will settle out over time and in many cases you can separate them from the solvent by using filter paper. Chalk in water and red blood cells in plasma are suspensions. Figures 15.3.2 and 15.3.3 show blood before and after it is separated into its component parts.



**FIGURE 15.3.2** On first glance, blood looks like a homogeneous mixture.



**FIGURE 15.3.3** Centrifuging blood separates it into clear layers, demonstrating that blood is actually a heterogeneous suspension. Here you can see the plasma on top and the red blood cells on the bottom.

A colloid is a mixture of particles that consists of smaller clusters of ions or molecules. These are evenly dispersed throughout the solvent and do not settle on standing. For example, milk is a colloid of a fat and an aqueous solution (Figure 15.3.4). Milk solids and fat are finely dispersed throughout the more aqueous components of the milk. Other common colloids include mayonnaise, paint and ink.



**FIGURE 15.3.4** Milk is a colloid. Small particles are dispersed throughout the liquid.

## Process of dissolving

The process of a substance dissolving in another substance is called **dissolution**. When the two substances are liquids, you can say they are **miscible**.

During dissolution the following processes occur.

- The particles of the solute are separated from one another.
- The particles of the solvent are separated from one another.
- The solute and solvent particles are attracted to each other.

For a solution to form, the solute particles must interact with the solvent molecules. The solute particles are surrounded by solvent molecules and carried throughout the solution.

## Forces involved in dissolving

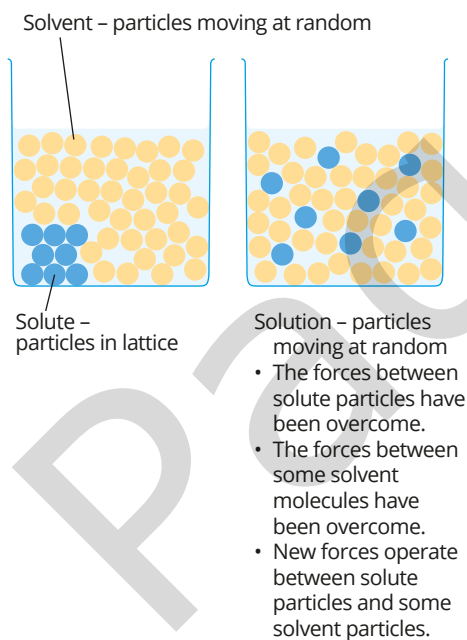
For a substance to dissolve, there must be a change in the way particles in the solute and solvent interact. This means you need to look at the forces of attraction that occur between the particles.

It is useful to think of three different forces of attraction when considering if and how a substance will dissolve in water. These are the forces:

- holding the particles of the substance (solute) together before it is added to the solvent
- holding the solvent molecules together. In water, these forces are hydrogen bonds
- that can form between the solute particles and solvent molecules if the substance dissolves.

For a substance to dissolve, the attractive forces that form between the solute and solvent particles must be sufficient to overcome the forces between the particles in the solute and the forces between the solvent molecules.

**i** For a substance to be soluble, the solute–solvent interactions must be stronger than the solute–solute and solvent–solvent interactions.



**FIGURE 15.3.5** The rearrangement of particles when a solute dissolves in a solvent. The solute particles separate and become evenly distributed in the solvent.

Figure 15.3.5 shows that as a solute dissolves, the solute particles separate and become evenly distributed in the solvent. If the attraction between the solute and solvent particles is not strong enough, then the substance will not readily dissolve.

You will recall from Chapter 12 that you can use a ‘like dissolves like’ rule to predict whether a substance is likely to dissolve in another substance. The general statement that a solvent will only dissolve ‘like’ solutes tells you that:

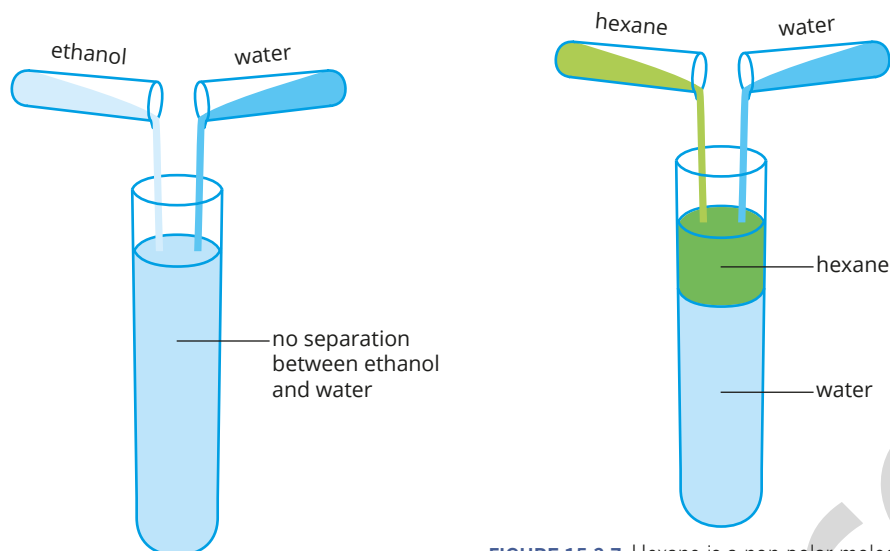
- **polar solvents** will generally dissolve substances consisting of polar molecules or charged ions, but will not dissolve solutes made up of non-polar molecules
- **non-polar solvents** can dissolve substances consisting of non-polar molecules, but will not dissolve ones with polar molecules or ions.

Oil and other non-polar molecular substances do not dissolve well in water because the only intermolecular forces in these substances are dispersion forces, whereas the strong intermolecular forces of hydrogen bonding exist between the water molecules. These hydrogen bonds between water molecules are much stronger than the dispersion forces that could occur between molecules of oil and water. As a result, the attraction between the water molecules cannot be overcome and the water molecules do not separate to form a solution with the oil molecules.

**i** Polar molecules have a permanent dipole, an uneven distribution of charge and are asymmetrical.

Non-polar molecules have bonds with an equal distribution of valence (outer shell) electrons therefore there is no charge on either end of the molecule and the molecule is symmetrical.

You can observe the 'like dissolves like' rule with the miscibility of different liquids. Figure 15.3.6 shows a solution of ethanol and water. The polar nature of the ethanol molecule means it readily dissolves in water, which is also polar. When mixed, a homogeneous solution is formed with no separation between solute and solvent.

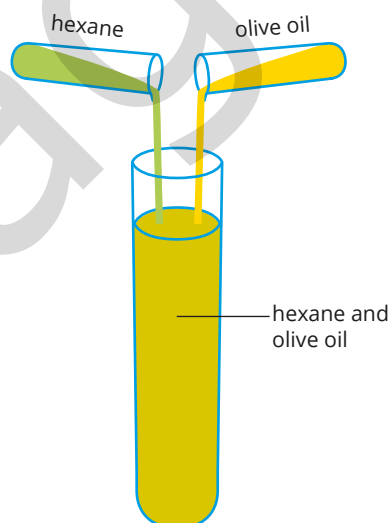


**FIGURE 15.3.6** The polar compound ethanol is completely miscible in water. A homogeneous solution is formed on mixing.

In contrast, hexane is immiscible in water. Hexane is composed of non-polar molecules that will not interact with the polar water molecules. Figure 15.3.7 shows that when hexane and water are mixed, two layers form and the less dense hexane sits on top of the water.

Like hexane, olive oil is also composed of non-polar molecules. When olive oil and hexane are mixed with each other, the 'like dissolves like' rule applies and the two non-polar liquids readily mix with each other to form a homogeneous solution (Figure 15.3.8).

**i** Use the 'like dissolves like' principle to predict if something will dissolve in a solvent.



**FIGURE 15.3.8** Hexane and olive oil are both non-polar molecular compounds and so are completely miscible.

## CHEMFILE

### Solubility of pharmaceutical drugs

Pharmaceutical drugs are usually carried through the blood to the place in a person's body where they need to act. The blood is mostly water so drugs need to dissolve in water to be transported. However, most drugs are organic molecules with low solubility in water. The force of attraction between water molecules and, often non-polar, drug molecules can be weak and so not overcome water-water molecular attractions and drug-drug molecular attractions.

To overcome this problem, a range of techniques are used to improve aqueous solubility of drugs, including physical and chemical modifications. Particle size reduction, a physical method of modification, involves producing the drugs in small particle size to increase their surface area which leads to more attractions between the drug and water molecules. Although pharmaceutical drugs are often large organic molecules, they often contain groups of atoms within their structure that can be modified to become positively or negatively charged—that is, they have some ionic character. These charged parts of the organic molecule form ion-dipole attractions with water molecules and so improve the drug's solubility. When chemical modifications of a potential drug are made there is testing to ensure it remains therapeutic and does not become toxic.

## 15.3 Review

### SUMMARY

- A solution is a homogeneous mixture in which molecules or ions are evenly dispersed throughout a solvent.
- Solutions in which water is the solvent are called aqueous solutions.
- A solution forms when the bonds between the solute and solvent particles are sufficiently strong to compete with attractive forces between the solute particles and between the solvent particles.
- 'Like dissolves like' is a rule that predicts that polar solutes will dissolve in polar solvents and non-polar solutes will dissolve in non-polar solvents.

### KEY QUESTIONS

- 1 Identify each of the following components of a glass of wine as solute, solvent or solution.
  - a water
  - b wine
  - c ethanol
  - d sugar
- 2 What is the one thing that all aqueous solutions have in common?
- 3 Classify the following substances as likely to be soluble or insoluble in water.
  - a ammonia ( $\text{NH}_3$ )
  - b oxygen gas ( $\text{O}_2$ )
  - c hydrogen chloride ( $\text{HCl}$ )
  - d methanol ( $\text{CH}_3\text{OH}$ )
  - e methane ( $\text{CH}_4$ )
  - f hydrogen fluoride ( $\text{HF}$ )
  - g carbon dioxide ( $\text{CO}_2$ )
- 4 Sodium chloride is very soluble in water. Explain what can be concluded about the forces present between the solute and solvent particles.
- 5 Match each of the following terms with the correct definition: dissolution, solution, solvent, solute, soluble, insoluble.
  - a capable of dissolving
  - b incapable of dissolving
  - c the minor component of a solution
  - d the process by which a substance is dissolved in a solvent
  - e a liquid mixture in which a solute is dispersed throughout a solvent
  - f the major component of a solution
- 6 Which one of the following molecules is likely to dissolve best in hexane, a non-polar solvent:  $\text{CH}_4$ ,  $\text{NH}_3$ ,  $\text{HF}$  or  $\text{H}_2\text{S}$ ? Explain your reasoning.



## 15.4 Water as a solvent of molecular substances

Most molecular substances are insoluble (or only very sparingly soluble) in water. However, some smaller molecules, such as ammonia ( $\text{NH}_3$ ), hydrogen chloride ( $\text{HCl}$ ) and sucrose ( $\text{C}_{12}\text{H}_{22}\text{O}_{11}$ ), dissolve well in water. Solutions in which a molecular substance is at least one of the solutes include brick cleaner, various liquid fertilisers and drinks such as wine and cordial (Figure 15.4.1).

There are two main ways in which molecular compounds dissolve well in water. In this section, you will explore both of these ways in detail.

### MOLECULAR COMPOUNDS THAT FORM HYDROGEN BONDS WITH WATER

One way a molecular compound might dissolve in water is if its molecules form hydrogen bonds with water molecules. An example of such a molecule is ethanol.

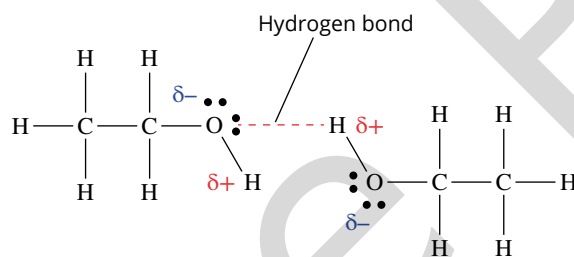
Ethanol ( $\text{C}_2\text{H}_5\text{OH}$ ) is a liquid at room temperature. Its molecules contain the polar  $-\text{OH}$  group, with lone pairs of electrons on the oxygen atom. The presence of the hydrogen atoms bonded to the electronegative oxygen atom allow an ethanol molecule to form hydrogen bonds.



**FIGURE 15.4.1** Lemon cordial is a solution that contains several dissolved molecular substances.

**i** Molecular compounds that can form hydrogen bonds are often soluble in water.

Figure 15.4.2 shows how hydrogen bonds form between molecules in pure ethanol.



**FIGURE 15.4.2** Hydrogen bonding in pure ethanol. The intermolecular hydrogen bond is formed between the lone pair electrons on the oxygen atom of one ethanol molecule and the electron-deficient hydrogen atom of an adjacent ethanol molecule.

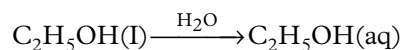
When ethanol is added to water, it dissolves. The two solutions are miscible. Figure 15.4.3 shows how hydrogen bonds form between the ethanol and surrounding water molecules.

Because the strength of the intermolecular forces between the solute and the solvent are similar, the two substances can readily interact with each other. Therefore, water and ethanol molecules mix freely with each other, held together in solution by hydrogen bonds.

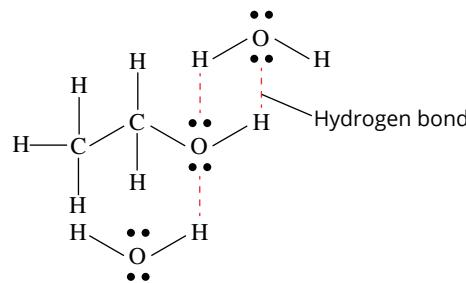
In summary, when ethanol dissolves in water, hydrogen bonds between:

- water molecules break
- ethanol molecules break
- ethanol molecules and water molecules form.

An equation for the dissolution of ethanol can be written to represent this process:



Note that the formula of water sits above the arrow. This is because there is no direct reaction between the water and the ethanol. The two substances simply mix together. No chemical change occurs; only the state symbol for ethanol is altered from (l) to (aq), indicating it is now dissolved in water.

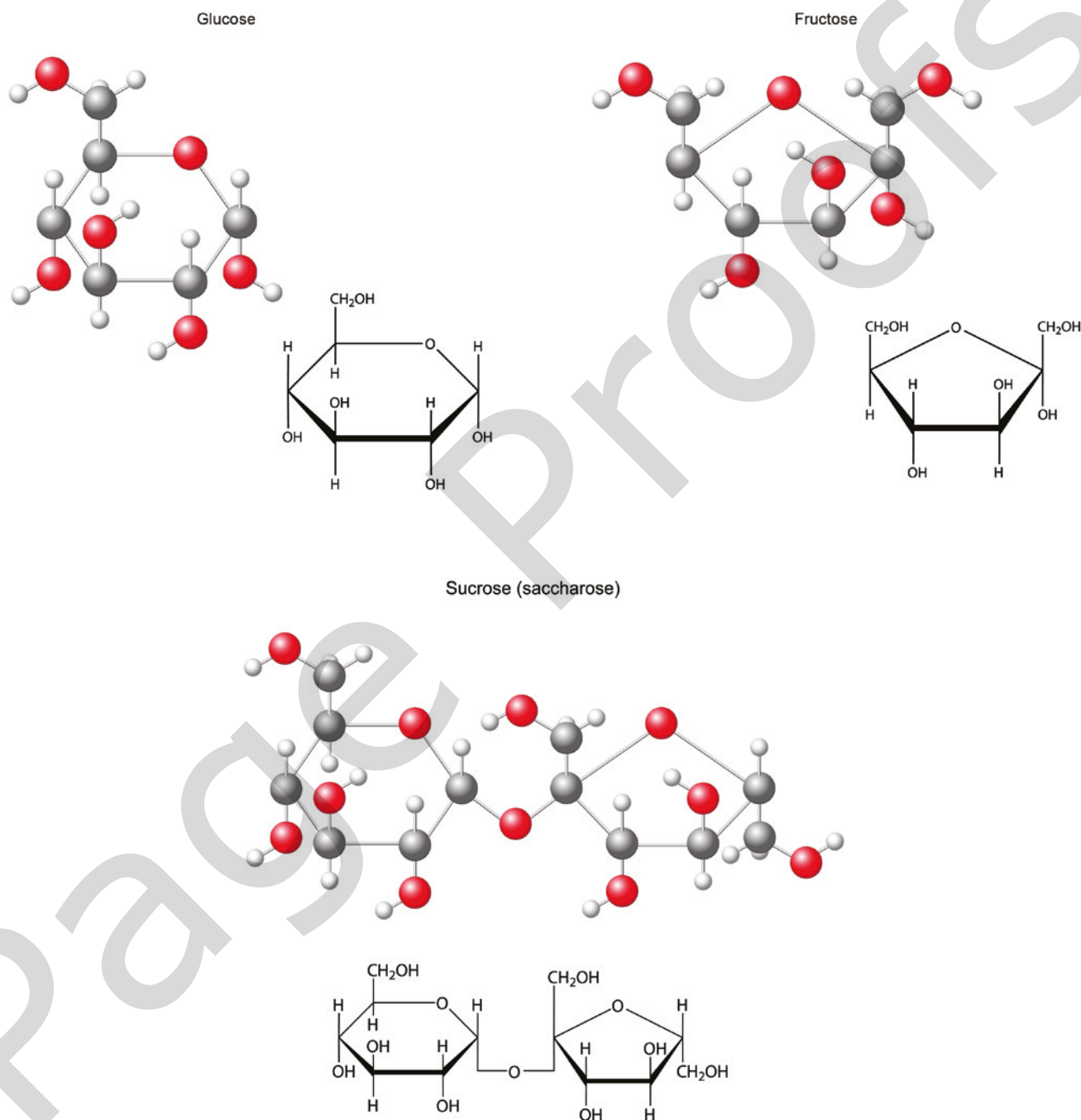


**FIGURE 15.4.3** Hydrogen bonding between ethanol and water.

**i** When a polar molecular substance dissolves in water by forming hydrogen bonds, the equation for dissolution places water above the arrow.

## Other polar compounds that form hydrogen bonds with water

Sugars, including glucose, fructose and sucrose, can all dissolve in water by forming hydrogen bonds. The structure of each of these sugars is shown in Figure 15.4.4. As with ethanol, their molecules also contain the polar  $\text{-OH}$  group.



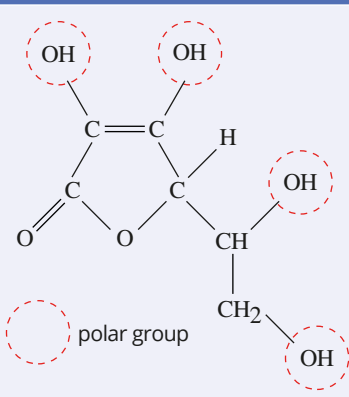
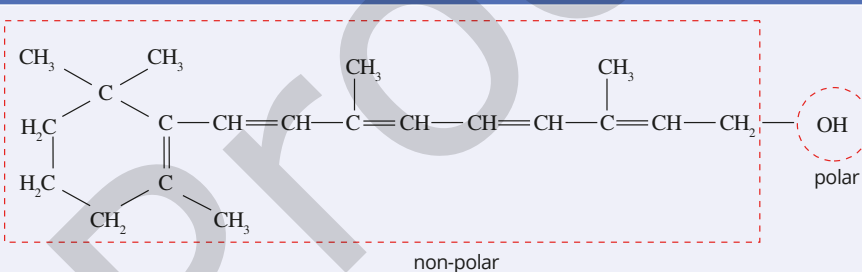
**FIGURE 15.4.4** These are the molecular structures of common sugars: glucose (the sugar found in our blood), fructose (fruit sugar) and sucrose (table sugar—the sugar used to sweeten tea and coffee). Each of these sugars can dissolve in water because of the presence of polar  $\text{-OH}$  groups.

The more polar the molecules of a molecular compound are, the more likely the compound is to dissolve in water. Some molecules have a polar section and a non-polar section. In general, the larger the non-polar section of the molecule, the less soluble it is in water.

Non-polar molecular substances do not have charged ends so there is no significant attraction to water molecules. The only force of attraction that exists between non-polar substances and water are weak dispersion forces, which are not strong enough to overcome the relatively strong hydrogen bonding between water molecules.

Table 15.4.1 compares the structures and properties of vitamin C and vitamin A. Both molecules contain the polar  $\text{-OH}$  group, but only vitamin C is soluble in water. The higher proportion of polar  $\text{-OH}$  groups on the vitamin C molecule allows it to form sufficient hydrogen bonds with water to overcome the strong attraction between water molecules, therefore dissolving to form a solution.

**TABLE 15.4.1** Comparison of the solubility and structures of vitamin C and vitamin A

|                         | Vitamin C                                                                                                                                                                                                                                | Vitamin A                                                                                                                                                                                                                                     |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Structure               |  <p>Vitamin C:</p> <ul style="list-style-type: none"> <li>contains four polar <math>\text{-OH}</math> groups</li> <li>molecule quite polar.</li> </ul> |  <p>Vitamin A:</p> <ul style="list-style-type: none"> <li>contains one polar <math>\text{-OH}</math> group</li> <li>molecule largely non-polar.</li> </ul> |
| Solubility              | Highly soluble in water; insoluble in fats                                                                                                                                                                                               | Soluble in fats, insoluble in water                                                                                                                                                                                                           |
| Biological significance | The high solubility of vitamin C means it is excreted in urine so it must form a regular part of the diet.                                                                                                                               | Vitamin A is not excreted in urine but is stored in body fat. The body can tolerate a diet low in vitamin A for a limited period.                                                                                                             |

## MOLECULAR COMPOUNDS THAT IONISE

Some compounds have molecules with one or more covalent bonds that are so polar they break when the compound is placed in water. Hydrogen chloride is such a compound. Hydrogen chloride ( $\text{HCl}$ ) is a gas at room temperature. Chlorine is much more electronegative than hydrogen so the  $\text{H-Cl}$  covalent bond is highly polar; the molecule forms a dipole.

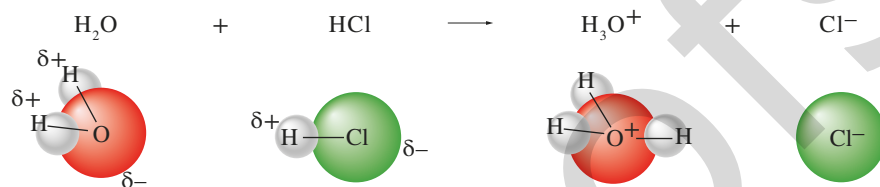
When hydrogen chloride is added to water, the hydrogen atom in  $\text{HCl}$  forms such a strong attraction to the oxygen atom in a water molecule that the  $\text{H-Cl}$  covalent bond breaks. The two electrons that made up the  $\text{H-Cl}$  covalent bond remain with the more electronegative  $\text{Cl}$  atom and the newly formed hydrogen ion ( $\text{H}^+$ ) joins the water molecule.

Figure 15.4.5 shows that when the hydrogen ion ( $\text{H}^+$ ) bonds to the water molecule, it forms a new ion known as the **hydronium ion** ( $\text{H}_3\text{O}^+$ ).

Since the Cl atom has gained an electron, it has a negative charge, forming a chloride ion ( $\text{Cl}^-$ ).

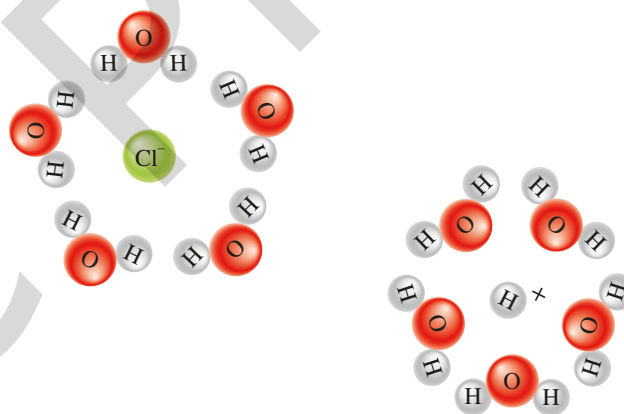
**i** Highly polar molecular compounds will form ions when dissolved in water.

The HCl molecule is split into smaller particles. The HCl is said to have become **ionised**; that is, it has undergone a process that has produced ions (Figure 15.4.5).



**FIGURE 15.4.5** The dipole–dipole attraction between the molecules of water and hydrogen chloride leads to the breaking of the polar covalent bond between the hydrogen and chlorine atoms. New ions, hydronium and chloride, are formed in a process called ionisation.

Figure 15.4.6 shows how the two ions produced in the reaction of HCl with water ( $\text{Cl}^-$  and  $\text{H}_3\text{O}^+$ ) become **hydrated**. The charged ions are surrounded by other polar water molecules. They are held in solution by **ion–dipole attractions**. (Ion–dipole attractions will be discussed in Section 15.5.)

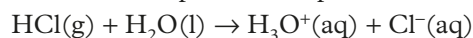


**FIGURE 15.4.6** When HCl molecules ionise in water, the newly formed ions are surrounded by adjacent water molecules. The charges of the ions interact strongly with the polar water molecules. For simplicity, the hydronium ion ( $\text{H}_3\text{O}^+$ ) is represented by the hydrogen ion ( $\text{H}^+$ ).

In summary, when hydrogen chloride dissolves in water:

- polar covalent bonds within hydrogen chloride molecules break, producing hydrogen ions ( $\text{H}^+$ ) and chloride ions ( $\text{Cl}^-$ )
- a covalent bond forms between each  $\text{H}^+$  and  $\text{H}_2\text{O}$  molecule, forming  $\text{H}_3\text{O}^+$  ions
- ion–dipole attractions form between the  $\text{H}_3\text{O}^+$  and  $\text{Cl}^-$  ions and polar water molecules.

An equation can be written to represent this process:



You should note two important points about this equation.

- Water is a reactant because there has been a rearrangement of atoms to form new substances.
- The aqueous state of the  $\text{H}_3\text{O}^+$  and  $\text{Cl}^-$  ions tells you that they are hydrated in solution.



## Other polar molecular compounds that ionise

Other compounds that dissolve in water by ionising include the common acids nitric acid ( $\text{HNO}_3$ ), sulfuric acid ( $\text{H}_2\text{SO}_4$ ) and ethanoic acid ( $\text{CH}_3\text{COOH}$ ).

Ammonia ( $\text{NH}_3$ ) is another polar molecule that can dissolve in water. Ammonia can dissolve by either forming hydrogen bonds with water or by reacting to form the ammonium ion ( $\text{NH}_4^+$ ). The process by which ammonia forms the ammonium ion will be looked at in detail in Chapter 17.

### CHEMFILE

#### Dissolved molecules in a fish tank

Many people keep fish at home in a small tank. Unfortunately, many pet fish do not live for quite as long as their owners hoped.

One of the biggest killers of aquarium fish is ammonia poisoning. Fish produce ammonia as a waste product. Ammonia is a polar molecule that readily dissolves in water, and is toxic to fish. Bacteria colonies living in a fish tank break down the ammonia produced by fish to other non-toxic products. In a healthy tank (Figure 15.4.7), a balance exists between ammonia production by fish and breakdown by bacteria.



**FIGURE 15.4.7** Fish in this tank produce the molecular compound ammonia ( $\text{NH}_3$ ) as a waste product. Bacteria living in the tank break down the toxic ammonia to harmless products.

## 15.4 Review

### SUMMARY

- Water is a good solvent for some polar molecular compounds.
- Some polar molecular compounds dissolve by forming hydrogen bonds with water.
- Some covalent molecular compounds are so polar that they dissolve in water by ionising to form hydrated ions.
- The table below summarises the two ways that molecular compounds dissolve in water.

| Type of solute                                 | Example of solute                         | Bond broken in the solute                                                    | Bonds formed                                                                                                                                                                                                                          | Equation                                                                                                                      |
|------------------------------------------------|-------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| polar covalent molecule that can hydrogen bond | ethanol, C <sub>2</sub> H <sub>5</sub> OH | hydrogen bonds broken between ethanol molecules                              | hydrogen bonds formed between ethanol and water molecules                                                                                                                                                                             | $\text{C}_2\text{H}_5\text{OH}(\text{l}) \xrightarrow{\text{H}_2\text{O}(\text{l})} \text{C}_2\text{H}_5\text{OH}(\text{aq})$ |
| polar molecules that ionise                    | hydrogen chloride, HCl                    | covalent bond broken between hydrogen and chlorine atoms in the HCl molecule | covalent bond formed between H <sup>+</sup> from HCl and oxygen atom in water molecule, forming H <sub>3</sub> O <sup>+</sup> ions. H <sub>3</sub> O <sup>+</sup> and Cl <sup>-</sup> ions form ion-dipole bonds with water molecules | $\text{HCl}(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{O}^+(\text{aq}) + \text{Cl}^-(\text{aq})$    |

### KEY QUESTIONS

- Which of the following substances are likely to dissolve in water by forming hydrogen bonds with water molecules.  
**A** ammonium sulfate ((NH<sub>4</sub>)<sub>2</sub>SO<sub>4</sub>)  
**B** methanol (CH<sub>3</sub>OH)  
**C** diamond (C)  
**D** hydrogen chloride (HCl)  
**E** ethylene glycol (HOCH<sub>2</sub>CH<sub>2</sub>OH)  
**F** sodium hydroxide (NaOH)  
**G** hydrogen gas (H<sub>2</sub>)
- Methanol (CH<sub>3</sub>OH) and glucose (C<sub>6</sub>H<sub>12</sub>O<sub>6</sub>) form hydrogen bonds with water. They will dissolve in water without ionising. Write chemical equations to represent the dissolving process for each of these compounds.
- With reference to the bonds that are broken and new bonds that are formed, describe how ethanol dissolves in water.
- Hydrogen iodide (HI) will ionise when it dissolves in water. The ionisation reaction is similar to that of HCl. Write a chemical equation to represent the dissolving process for this compound.
- Hydrogen chloride dissolves in water by ionisation. Using the terms provided, complete the following sentences to describe the process of how ionisation occurs. Some terms may be used more than once.  
 ion-dipole bonds                      hydrogen bonds  
 covalent bonds                        hydronium  
 ionises                                    H<sup>+</sup>  
 \_\_\_\_\_ within hydrogen chloride molecules are broken.  
 \_\_\_\_\_ between water molecules are broken.  
 The HCl \_\_\_\_\_ and produces Cl<sup>-</sup> and H<sup>+</sup> ions.  
 \_\_\_\_\_ form between \_\_\_\_\_ ions and water to produce \_\_\_\_\_ ions. \_\_\_\_\_ form between the Cl<sup>-</sup> and H<sub>3</sub>O<sup>+</sup> ions and polar water molecules.

## 15.5 Water as a solvent of ionic compounds

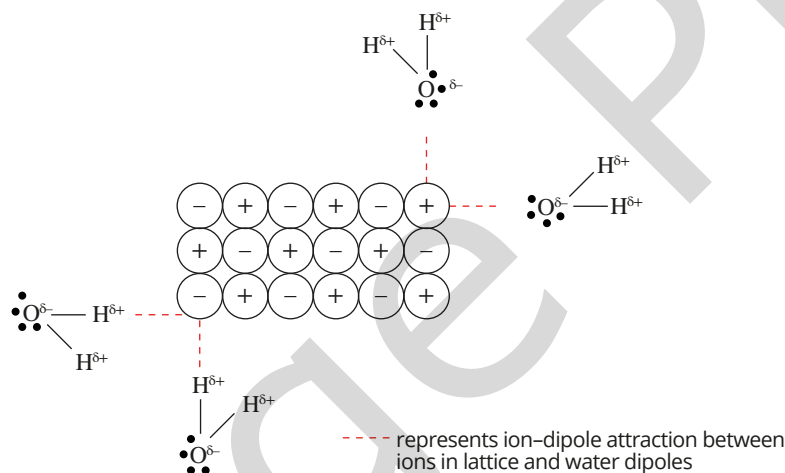
Sports drinks are advertised to athletes as a way to replace the electrolytes lost in sweat during exercise (Figure 15.5.1). The electrolytes are dissolved ionic substances such as sodium chloride and potassium phosphate.

Many ionic substances are soluble in water. However, not all ionic substances dissolve readily in water. In this section, you will look at how ionic substances dissolve and learn how to predict whether an ionic compound will be soluble in water.

### DISSOLUTION OF AN IONIC LATTICE IN WATER

Many ionic compounds dissolve readily in water. Sodium chloride is a typical ionic compound that exists as a solid at room temperature. In Figure 15.5.2, you can see the arrangement of sodium cations ( $\text{Na}^+$ ) and chloride anions ( $\text{Cl}^-$ ) in a three-dimensional ionic lattice. The ions are held together by strong electrostatic forces between the positive and negative charges of the ions.

When an ionic compound such as sodium chloride is added to water, the positive ends of the water molecules are attracted to the negatively charged chloride ions and the negative ends of the water molecules are attracted to the positively charged sodium ions (Figure 15.5.3). The attraction between an ion and a polar molecule such as water is described as an ion–dipole attraction.



**FIGURE 15.5.3** Electrostatic attraction occurs between the negative ions in a NaCl lattice and the hydrogen atoms in polar water molecules. Electrostatic attraction also occurs between the positive ions in the lattice and the oxygen atoms in water molecules.

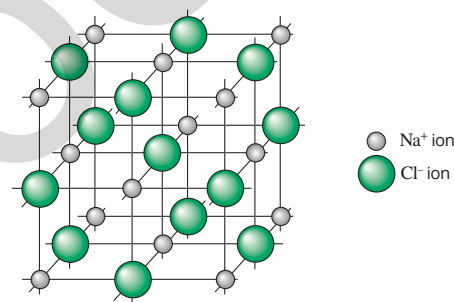
Water molecules are in a continuous state of random motion. If the ion–dipole attractions between the ions and the water molecules are strong enough, the water molecules can pull the sodium and chloride ions on the outer part of the crystal out of the lattice and into the surrounding solution.

Sodium ions and chloride ions pulled out of the lattice become surrounded by water molecules. These ions are said to be hydrated. Water molecules are arranged around the ions as shown in Figure 15.5.4. Note the different arrangements of the water molecules around the positive and negative ions. The hydrogen atoms in the water molecule are more positive so they are orientated towards the negative chloride ion. The positive sodium ion is surrounded by the more negative oxygen atom of the water molecules.

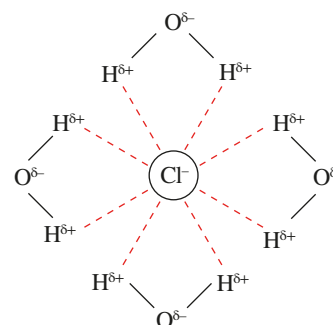
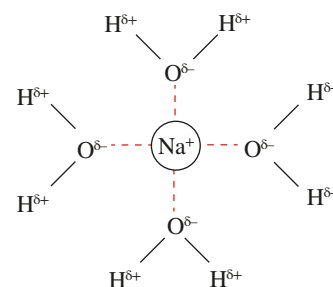
The process of separating positive and negative ions from a solid ionic compound to form hydrated ions when an ionic compound dissolves in water is called **dissociation**.



**FIGURE 15.5.1** Sports drinks are used by athletes to replace water and dissolved ionic solutes.



**FIGURE 15.5.2** A representation of the crystal lattice of the ionic compound sodium chloride.



--- represents ion–dipole interaction

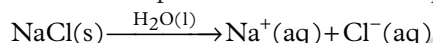
**FIGURE 15.5.4** Ion–dipole attraction between the ions and adjacent water molecules to form hydrated sodium and chloride ions.

Although the ionic bonds within the lattice are strong, the ions can be pulled away from the lattice by the interactions of many water molecules.

In summary, when sodium chloride dissolves in water:

- ionic bonds within the sodium chloride lattice are broken
- hydrogen bonds between water molecules are broken
- ion–dipole attractions form between ions and polar water molecules.

An equation can be written to represent the dissociation process:



Note that the formula of water sits above the arrow. This is because there is no direct reaction between the water and the sodium chloride. No chemical change occurs; only the state symbol for sodium chloride is altered from (s) to (aq), indicating it is now dissolved in water. (You may omit the  $\text{H}_2\text{O}$  from this equation if you wish.)

It is important to note that dissociation of ionic compounds is simply freeing ions from the lattice so that they can move throughout the solution. This is different from ionisation of molecular compounds where new ions are formed by the reaction of the molecule with water.

**i** Ionic substances dissolve by dissociation. Ion–dipole bonds are formed between the ions and water molecules.



**FIGURE 15.5.5** Limestone caves are formed by limestone (calcium carbonate) dissolving over very long periods of time. This photograph is of a limestone cave at Loch Ard Gorge, Victoria.

## INSOLUBLE IONIC COMPOUNDS

Not all ionic compounds are soluble in water. For example, limestone ( $\text{CaCO}_3$ ) is almost completely insoluble in water. Limestone caves such as those seen in Figure 15.5.5 are formed over a long period of time as  $\text{CaCO}_3$  is dissolved and redeposited. The ionic compound  $\text{Ca}_3(\text{PO}_4)_2 \cdot \text{Ca}(\text{OH})_2$ , which gives strength to bones and teeth, is also (fortunately) insoluble in water.

Insoluble ionic compounds do not dissolve in water because the energy required to separate the ions from the lattice is greater than the energy released when the ions are hydrated.

Although substances are usually described as ‘soluble’ or ‘insoluble’, this is a generalisation. Substances that are described as ‘insoluble’ tend to dissolve very slightly. Those that are described as ‘soluble’ dissolve to varying extents.

**i** Substances are rarely ever completely ‘soluble’ or ‘insoluble’. The solubility of substances can be considered as a scale.

## The SNAPE rule

A handy way to remember many of the soluble salts is with the initials SNAPE. Salts that contain one or more of the following ions are soluble:

- **S**odium ( $\text{Na}^+$ )
- **N**itrate ( $\text{NO}_3^-$ )
- **A**mmonium ( $\text{NH}_4^+$ )
- **P**otassium ( $\text{K}^+$ )
- **E**thanoate ( $\text{CH}_3\text{COO}^-$ )

## Solubility tables

A **solubility table** can be used to determine whether common ionic compounds are soluble in water. Anions and cations are listed with indications of their solubility in different compounds. Table 15.5.1 is a sample solubility table listing ionic compounds that are soluble in water. Note that for some ions there are exceptions. Insoluble compounds are given in Table 15.5.2.



**TABLE 15.5.1** Relative solubilities of soluble ionic compounds

| Soluble in water (>0.1 mol dissolves per L at 25°C)                                       | Exceptions: insoluble (<0.1 mol dissolves per L at 25°C)      | Exceptions: slightly soluble (0.01–0.1 mol dissolves per L at 25°C) |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------|
| most chlorides ( $\text{Cl}^-$ ), bromides ( $\text{Br}^-$ ) and iodides ( $\text{I}^-$ ) | $\text{AgCl}$ , $\text{AgBr}$ , $\text{AgI}$ , $\text{PbI}_2$ | $\text{PbCl}_2$ , $\text{PbBr}_2$                                   |
| all nitrates ( $\text{NO}_3^-$ )                                                          | no exceptions                                                 | no exceptions                                                       |
| all ammonium ( $\text{NH}_4^+$ ) salts                                                    | no exceptions                                                 | no exceptions                                                       |
| all sodium ( $\text{Na}^+$ ) and potassium ( $\text{K}^+$ ) salts                         | no exceptions                                                 | no exceptions                                                       |
| all ethanoates ( $\text{CH}_3\text{COO}^-$ )                                              | no exceptions                                                 | no exceptions                                                       |
| most sulfates ( $\text{SO}_4^{2-}$ )                                                      | $\text{SrSO}_4$ , $\text{BaSO}_4$ , $\text{PbSO}_4$           | $\text{CaSO}_4$ , $\text{Ag}_2\text{SO}_4$                          |

**TABLE 15.5.2** Relative solubilities of insoluble ionic compounds

| Insoluble in water                     | Exceptions: soluble                                                                                        | Exceptions: slightly soluble                        |
|----------------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| most hydroxides ( $\text{OH}^-$ )      | $\text{NaOH}$ , $\text{KOH}$ , $\text{Ba}(\text{OH})_2$ , $\text{NH}_4\text{OH}^*$ , $\text{AgOH}^\dagger$ | $\text{Ca}(\text{OH})_2$ , $\text{Sr}(\text{OH})_2$ |
| most carbonates ( $\text{CO}_3^{2-}$ ) | $\text{Na}_2\text{CO}_3$ , $\text{K}_2\text{CO}_3$ , $(\text{NH}_4)_2\text{CO}_3$                          | no exceptions                                       |
| most phosphates ( $\text{PO}_4^{3-}$ ) | $\text{Na}_3\text{PO}_4$ , $\text{K}_3\text{PO}_4$ , $(\text{NH}_4)_3\text{PO}_4$                          | no exceptions                                       |
| most sulfides ( $\text{S}^{2-}$ )      | $\text{Na}_2\text{S}$ , $\text{K}_2\text{S}$ , $(\text{NH}_4)_2\text{S}$                                   | no exceptions                                       |

\* $\text{NH}_4\text{OH}$  does not exist in significant amounts in an ammonia solution. Ammonium and hydroxide ions readily combine to form ammonia and water.

† $\text{AgOH}$  readily decomposes to form a precipitate of silver oxide and water.

## Using a solubility table to predict solubility

The solubility tables provided as Tables 15.5.1 and 15.5.2 can be used to predict whether a particular ionic substance will be soluble, slightly soluble or insoluble in water.

Worked example 15.5.1 shows you a systematic way of using a solubility table to determine whether an ionic compound will be soluble or insoluble in water.

### Worked example 15.5.1

#### DETERMINING IF IONIC COMPOUNDS ARE SOLUBLE OR INSOLUBLE IN WATER

|                                                                                                                                                    |                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Is barium sulfide ( $\text{BaS}$ ) soluble or insoluble in water? You will need to refer to the solubility tables above to complete this question. |                                                                                                                  |
| <b>Thinking</b>                                                                                                                                    | <b>Working</b>                                                                                                   |
| Identify the ions that are present in the ionic compound.                                                                                          | Barium ( $\text{Ba}^{2+}$ ) and sulfide ( $\text{S}^{2-}$ )                                                      |
| Check the solubility tables to see if compounds containing the cation are usually soluble or insoluble in water.                                   | Barium ions do not appear in the table. This is no help, so look for the other ion.                              |
| Check the solubility tables to see if compounds containing the anion are usually soluble or insoluble in water.                                    | Compounds containing sulfide ions are usually insoluble in water. So, barium sulfide will be insoluble in water. |

### Worked example: Try yourself 15.5.1

#### DETERMINING IF IONIC COMPOUNDS ARE SOLUBLE OR INSOLUBLE IN WATER

Is ammonium phosphate ( $(\text{NH}_4)_3\text{PO}_4$ ) soluble or insoluble in water? You will need to refer to the solubility tables above to complete this question.

## CHEMFILE

### The Dead Sea

The Dead Sea, a salt lake in the Middle East, is one of the saltiest bodies of water on Earth. The concentration of dissolved ionic compounds, including salt, in the water makes the Dead Sea roughly 8.6 times saltier than the ocean. This prevents fish and aquatic plants from living in it. There is so much dissolved salt in the water that people can easily float in the Dead Sea due to natural buoyancy; the density of the salt water is greater than the density of the human body (Figure 15.5.6).



**FIGURE 15.5.6** The Dead Sea contains so many dissolved ions it is easy to float.

## 15.5 Review

### SUMMARY

- Soluble ionic compounds dissociate in water to form hydrated ions.
- In a hydrated ion, water molecules are attracted to the central ion by ion–dipole attractions.
- The table below summarises the way soluble ionic compounds dissolve in water.
- A solubility table can be used to determine whether a particular ionic compound is likely to be soluble or insoluble in water.

| Type of solute          | Example                | Bond broken in the solute                                | Bonds formed with water                                                   | Equation                                                                                             |
|-------------------------|------------------------|----------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| soluble ionic compounds | sodium chloride (NaCl) | ionic bonds between $\text{Na}^+$ and $\text{Cl}^-$ ions | ion–dipole attractions between dissociated ions and polar water molecules | $\text{NaCl(s)} \xrightarrow{\text{H}_2\text{O(l)}} \text{Na}^+(\text{aq}) + \text{Cl}^-(\text{aq})$ |

### KEY QUESTIONS

- Sodium nitrate ( $\text{NaNO}_3$ ) and calcium hydroxide ( $\text{Ca(OH)}_2$ ) will both dissociate when they dissolve in water. Write chemical equations to represent the dissolving process for each of these compounds.
- Redraw the representations of  $\text{Na}^+$  and  $\text{Cl}^-$  ions below, then show the correct orientation of water molecules around hydrated sodium and chloride ions. Use a V-shape to represent the shape of water molecules.
 

$\text{Na}^+$

$\text{Cl}^-$
- Which of the following substances would you expect to be soluble in water? Refer to Table 15.5.1 and Table 15.5.2 on page XX to complete this question.
  - sodium carbonate
  - lead(II) nitrate
  - magnesium carbonate
  - ammonium sulfate
  - iron(II) sulfate
  - magnesium phosphate
  - zinc carbonate
  - sodium sulfide
  - silver chloride
  - barium sulfate
- Which of the following compounds would you expect to be insoluble in water? Refer to Table 15.5.1 and Table 15.5.2 on page XX to complete this question.
  - silver carbonate
  - zinc nitrate
  - copper(II) carbonate
  - silver chloride
  - lead(II) bromide
  - magnesium hydroxide
  - barium nitrate
  - aluminium sulfide
- Write the formulae for the ions produced when these compounds dissolve in water.
  - sodium carbonate
  - calcium nitrate
  - potassium bromide
  - iron(III) sulfate
  - copper(II) chloride
- Suggest reasons for each of the following.
  - Concentrated deposits of nitrate compounds are only found in desert regions.
  - The sea is a rich source of sodium, chloride and sulfate ions.

## 15.6 Solubility

The solubility of a substance is a measure of how much of the substance will dissolve in a given amount of a **solvent**. For example, glucose is very soluble in water and is used by your body as a readily available energy source. Most rocks are made of minerals that are insoluble in water, but limestone (which contains calcium carbonate) is slightly soluble and so caves form in areas of limestone rock over long periods.

In this section, you will learn how to measure the solubility of some compounds in water and how to predict whether a compound will be soluble or not.

### DEFINITION OF SOLUBILITY

In chemistry, the term **solubility** has a specific meaning. It refers to the maximum amount of a **solute** (substance that is dissolved) that can be dissolved in a given quantity of a solvent (the common quantity often used is 100 g) at a certain temperature. Table 15.6.1 gives the solubility of some common substances in 100 g of water at 18°C.

**TABLE 15.6.1** Solubility of solutes at 18°C

| Solute                        | Solubility (g per 100 g of water) |
|-------------------------------|-----------------------------------|
| table sugar (sucrose)         | 200                               |
| table salt (sodium chloride)  | 35                                |
| limestone (calcium carbonate) | 0.0013                            |

### Different kinds of solutions

There are three terms that can be used to describe the different solutions that result from dissolving solutes in solvents. They are:

- saturated
- unsaturated
- supersaturated.

A **saturated solution** is one in which no more solute can be dissolved at a particular temperature.

An **unsaturated solution** contains less solute than is needed to make the solution saturated. Unsaturated solutions can dissolve more solute.

A **supersaturated solution** is an unstable solution that contains more dissolved solute than a saturated solution. If this type of solution is disturbed, some of the solute will separate from the solvent as a solid.

Figure 15.6.1 shows a supersaturated solution of sodium ethanoate. Supersaturated sodium ethanoate is prepared by cooling a saturated solution very carefully so that solid crystals do not form. Adding a small **seed crystal** to the supersaturated solution causes the solute to **crystallise** (form solid crystals) so that a saturated solution remains.

**i** Saturated solutions are those in which no more solute can be dissolved in a solvent (usually water) at a particular temperature.



**FIGURE 15.6.1** Crystals of sodium ethanoate form after a seed crystal is added to a supersaturated solution of the compound.

## SOLUBILITY CURVES

The solubility of many substances changes significantly as the temperature changes. For example, you can dissolve more chocolate powder in hot milk than in cold milk. For most solids, increasing the temperature increases the solubility in a liquid. This is because at higher temperatures, both the solute and solvent have more energy to overcome the forces of attraction that hold the particles together in the solid.

The relationship between solubility and temperature can be represented by a solubility curve, as shown in Figure 15.6.2.

**Solubility curves** show the solubility of a substance as a function of temperature. For the solubility curves featured in Figure 15.6.2, note that:

- each point on a curve represents the maximum amount of the solute that can be dissolved in 100 g of water at a particular temperature. Therefore, each point on the curve represents a saturated solution
- any point below a curve represents an unsaturated solution
- any point above a curve represents a supersaturated solution
- for most solids, as temperature increases, the solubility increases.

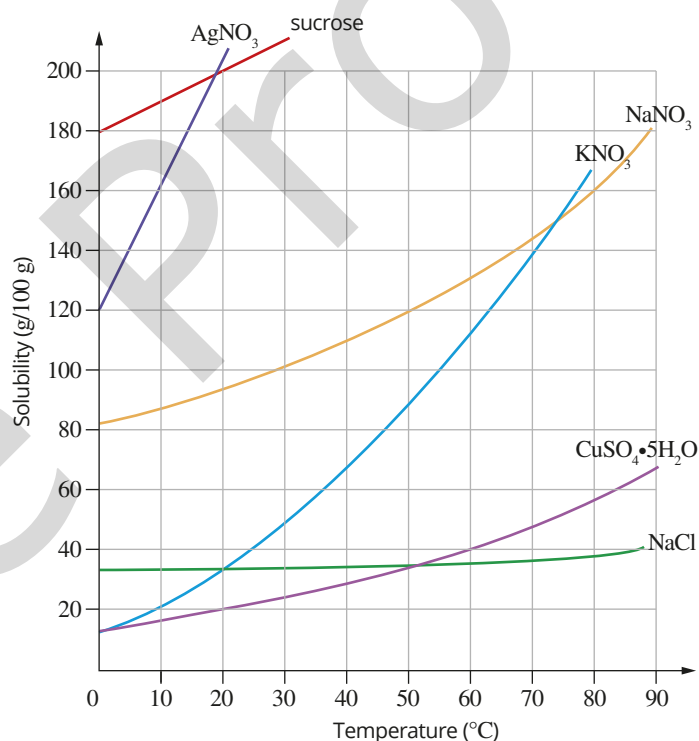


FIGURE 15.6.2 Solubility curves for some common chemicals



## Worked example 15.6.1

### SOLUBILITY CURVE CALCULATIONS

Use Figure 15.6.2 on page XXX to find how many grams of sodium nitrate ( $\text{NaNO}_3$ ) will dissolve in 100 g of water at 50°C.

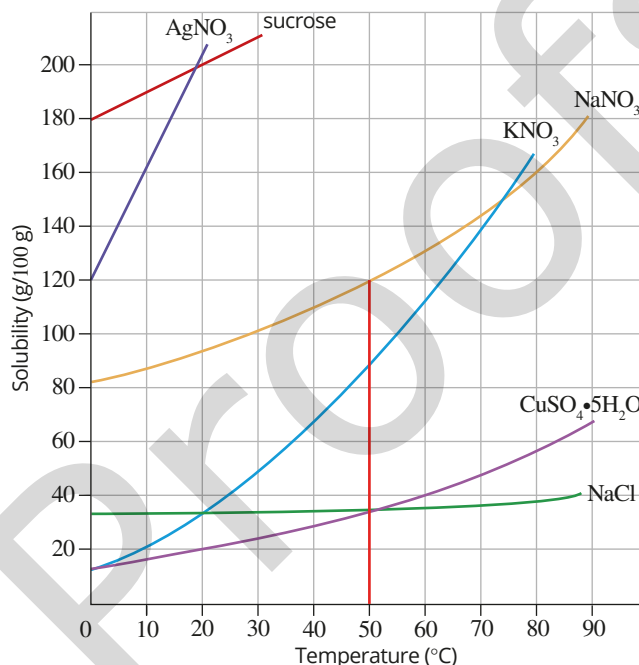
#### Thinking

Points on a solubility curve show the maximum amount of the solute that will dissolve in 100 g water at a given temperature.

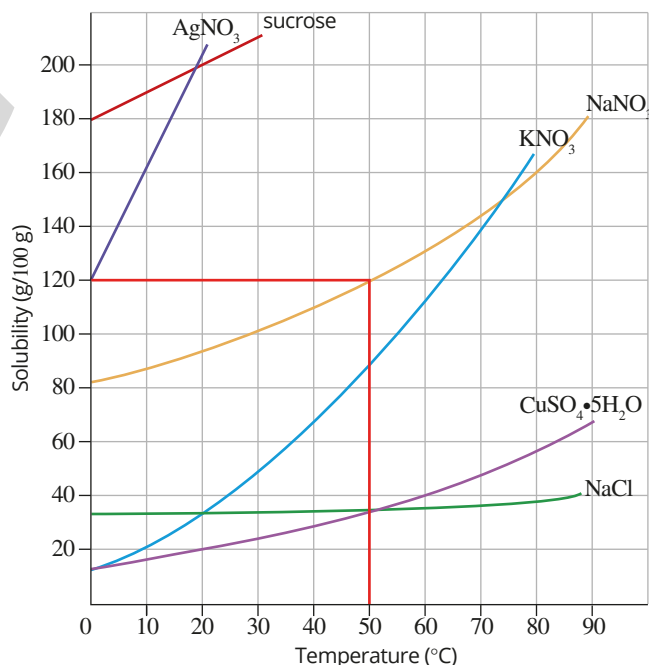
Draw a vertical line from the required temperature on the horizontal axis to intersect with the appropriate solubility curve.

#### Working

Draw a vertical line from 50°C on the horizontal axis to intersect with the solubility curve for  $\text{NaNO}_3$ .



Draw a horizontal line from the intersection point of the vertical line drawn in the previous step. The point where this horizontal line intersects with the vertical axis will give the mass of dissolved substance.



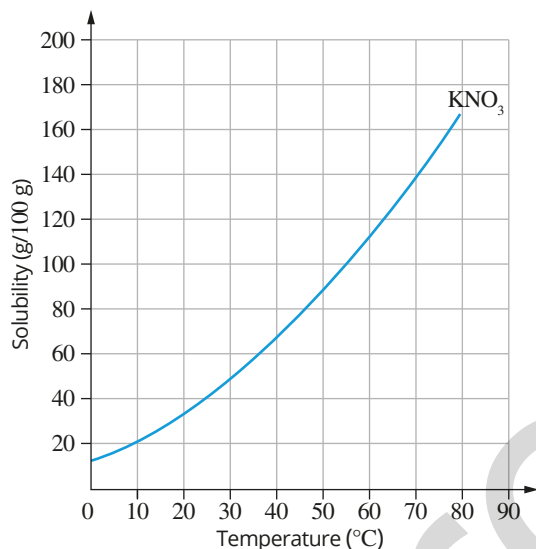
Read the solubility from the graph.

The horizontal line intersects with the vertical axis at 120 g. Therefore, the mass of  $\text{NaNO}_3$  that will dissolve in 100 g of water at 50°C is 120 g.

### Worked example: Try yourself 15.6.1

#### SOLUBILITY CURVE CALCULATIONS

Use the solubility curve below to find how many grams of potassium nitrate ( $\text{KNO}_3$ ) will dissolve in 100 g of water at  $70^\circ\text{C}$ .



### More calculations using solubility curves

Solubility curves allow you to directly calculate the mass of a compound that will dissolve in 100 g water at different temperatures. It is also possible to use the curves to calculate solubilities of compounds in quantities of water other than 100 g. Worked example 15.6.2 shows you how.

### Worked example 15.6.2

#### SOLUBILITY CURVES

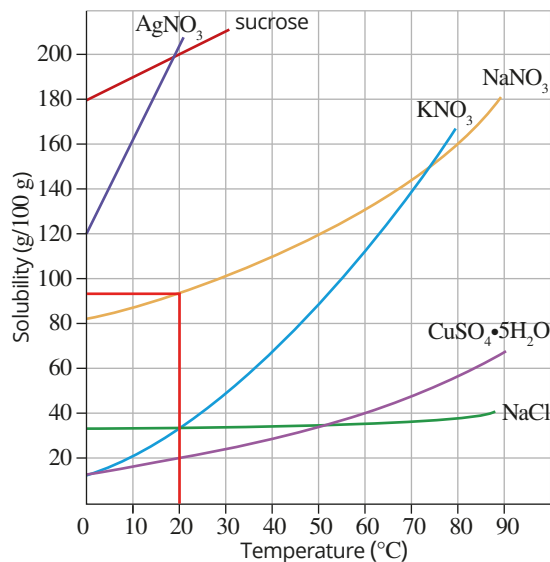
An 80 g sample of sodium nitrate ( $\text{NaNO}_3$ ) is added to 200 mL of  $\text{H}_2\text{O}$  at  $20^\circ\text{C}$ . Use Figure 15.6.2 on page XXX to calculate how much more  $\text{NaNO}_3$  must be added to make the solution saturated with  $\text{NaNO}_3$  at  $20^\circ\text{C}$ .

##### Thinking

Use the solubility curve to find the mass of solute in a saturated solution of 100 g of  $\text{H}_2\text{O}$  at the required temperature.

##### Working

Draw a line from  $20^\circ\text{C}$  on the horizontal axis to the solubility curve for  $\text{NaNO}_3$  and find the corresponding value on the vertical axis. The value is 92 g.



|                                                                                                                                                                                     |                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use the amount of solute that will dissolve in 100 g of H <sub>2</sub> O to find the mass of solute to make a saturated solution in the mass of H <sub>2</sub> O for this question. | The density of water is 1.0 g mL <sup>-1</sup> , so 200 mL of water will weigh 200 g. So twice the mass of solute can dissolve in 200 g of water as in 100 g.<br>$m(\text{NaNO}_3) = 2 \times 92$ $= 184 \text{ g}$ |
| To find out how much extra solute you need to add, find the difference between the mass of solute needed to make a saturated solution and how much has already been added.          | 80 g of NaNO <sub>3</sub> has already been added to 200 g H <sub>2</sub> O. So the extra mass of NaNO <sub>3</sub> needed:<br>$= 184 - 80$ $= 104 \text{ g}$                                                        |

### Worked example: Try yourself 15.6.2

#### SOLUBILITY CURVES

A 120 g sample of sodium nitrate (NaNO<sub>3</sub>) is added to 300 mL of H<sub>2</sub>O at 40°C. Use Figure 15.1.2 on page XXX to calculate how much more NaNO<sub>3</sub> must be added to make the solution saturated with NaNO<sub>3</sub> at 40°C.

#### Making your own solubility curve

Solubility curves can be developed relatively easily in a school laboratory. For example, a solubility curve for potassium nitrate in water can be derived in a group activity by carrying out the steps shown in Figure 15.6.3.

#### PROCESS OF CRYSTALLISATION

##### Methods of crystallisation

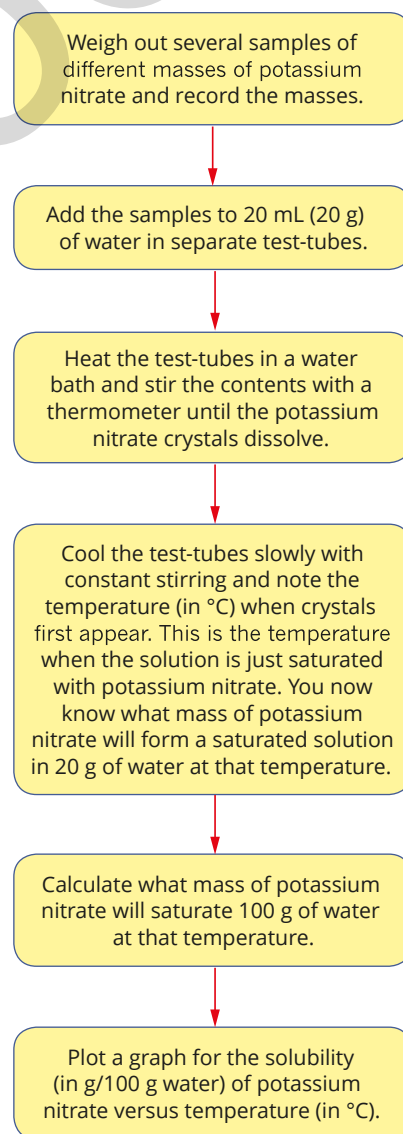
**Crystallisation** from solutions occurs when an unsaturated solution becomes saturated and crystals form. There are two main ways in which this can happen.

Cooling a solution may reduce the solubility of a dissolved solute to the point where not all of the substance present is soluble. An example of this occurs when copper(II) sulfate crystals form after a hot solution of the compound is cooled. At lower temperatures, less of the compound can dissolve in the water and a point will be reached when the solution becomes saturated. Further cooling will result in crystals being formed.

Alternatively, crystallisation can occur as a result of the evaporation of solvent from a solution. For example, salt forms as water evaporates from seawater. This process produced the crystals of salt seen in Figure 15.6.4.



**FIGURE 15.6.4** White crystals of salt are left behind at the edge of a rock pool where seawater has evaporated.



**FIGURE 15.6.3** Steps for constructing a solubility curve for potassium nitrate in water

## CHEMFILE

### Salt

In Australia, salt is made from seawater in salt pans (Figure 15.6.5). The concentration of sodium chloride in seawater is approximately  $35\text{ g L}^{-1}$ . Seawater is released into shallow ponds and the water evaporates due to heat from the Sun. As the volume of water decreases, the concentration of salt increases. Eventually, the solution becomes saturated and the salt begins to crystallise out. Australia's hot climate makes it very suitable for producing salt by evaporation, and it is a major exporter of salt.



**FIGURE 15.6.5** Salt is produced by the evaporation of water from seawater.



**FIGURE 15.6.6** Large crystals of copper(II) sulfate formed by slowly cooling a saturated solution of the compound.

## Variables affecting crystal growth

Several factors affect the formation of crystals from solutions. Some of these factors can affect the size of crystals formed.

### Rate of cooling of a solution or molten compound

The rate of temperature change during cooling can affect the size of the crystals that are formed.

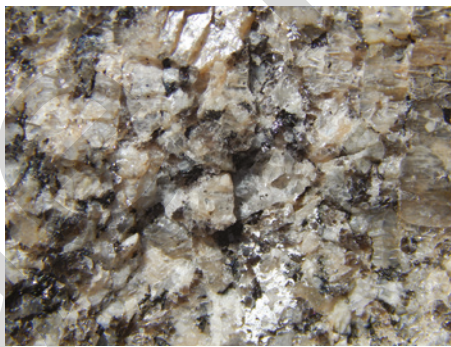
- Slow cooling generally produces large crystals.
- Rapid cooling produces smaller crystals.

This is true whether crystals are formed from a solution or as a molten compound cools below its melting point.

You can investigate this phenomenon in a laboratory. For example, a saturated solution of copper(II) sulfate can be cooled at different rates. Some of the solution could be cooled quickly by placing it in a refrigerator. Another sample could be cooled slowly by allowing it to sit in an insulated dish. The slower cooling results in larger crystals, as shown in the Figure 15.6.6.

Igneous rocks illustrate the effects of cooling rates on crystal size. The molten magma from the Earth's interior that forms these rocks can cool down quickly on the Earth's surface during volcanic activity, or very slowly underground. Figure 15.6.7 shows a section of granite, a rock that forms when magma cools below the Earth's surface. As it cools below the surface, the rate of temperature change is slow so granite contains large crystals.

On the other hand, basalt is a rock that forms when magma flows out from a vent in the Earth and over surrounding land. This results in rapid cooling of the magma and formation of small crystals. The basalt plains around Melbourne and much of southwestern Victoria were formed in this way (Figure 15.6.8). The bluestone paving and kerb stones used for many streets and lanes in the city and older suburbs of Melbourne were cut from these basalt deposits.



**FIGURE 15.6.7** Large crystals (or grains) in granite.



**FIGURE 15.6.8** The rock columns at the Organ Pipes National Park near Melbourne are basalt extrusions.

### Rate of evaporation of solvent

If crystals are formed by evaporation, then the rate of evaporation of the solvent will affect the size of the crystals. Generally, a faster rate of evaporation results in smaller crystals. This can be investigated in the laboratory by placing solutions of a chemical in different situations where the rate of evaporation of the solvent varies, for example in a cupboard and on a windowsill.

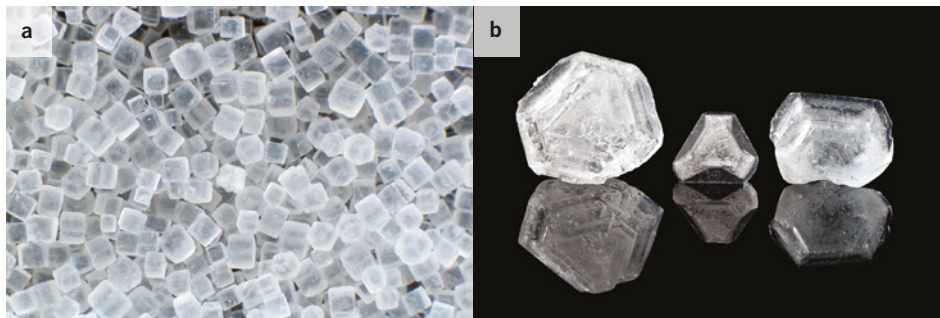
### Nucleation

Crystallisation occurs more rapidly if a 'nucleus' is present in the solution on which the new crystals can form. A nucleus can be provided by adding a single seed crystal of the compound itself. You saw an example of this in Figure 15.6.1 on page XX. Even a small solid impurity in the solution, such as a speck of dust or a scratch on the container wall can act as a nucleus.



### Nature of the compound

Different compounds produce different-shaped crystals. Figure 15.6.9 shows two different-shaped crystals. Sodium chloride crystals are cubic in shape; the crystals of alum (potassium aluminium sulfate,  $\text{KAl}(\text{SO}_4)_2$ ) are octahedral.



**FIGURE 15.6.9** (a) Cubic crystals of table salt (sodium chloride) and (b) octahedral crystals of alum,  $\text{KAl}(\text{SO}_4)_2$

**i** Crystallisation occurs when an unsaturated solution becomes saturated and solute crystals start to form. This can be due to evaporation of solvent, cooling of the solution or the addition of a seed crystal.

### Calculations involving crystallisation

You can predict how much of a compound will crystallise out of a solution as temperature decreases by using a solubility curve. Worked example 15.6.3 shows how this can be done.

#### Worked example 15.6.3

##### CRYSTALLISATION

50g of potassium nitrate ( $\text{KNO}_3$ ) is dissolved in 100g water at  $40^\circ\text{C}$ . What mass of  $\text{KNO}_3$  crystals will form if the temperature is reduced to  $20^\circ\text{C}$ ? You will need to refer to Figure 15.6.2 on page XX to complete this question.

| Thinking                                                                                           | Working                                                                                           |
|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Identify the mass of solute dissolved in the original solution.                                    | Mass of $\text{KNO}_3$ in solution = 50g                                                          |
| Find the maximum mass of solute that will remain dissolved in 100g water at the final temperature. | The solubility curve shows that the maximum mass that will dissolve at $20^\circ\text{C}$ is 32g. |
| Calculate the mass of solute crystals that will form in the solution at the final temperature.     | Mass of crystals formed = original mass – remaining mass<br>$= 50 - 32$<br>$= 18\text{g}$         |

#### Worked example: Try yourself 15.6.3

##### CRYSTALLISATION

200g of sucrose is dissolved in 100g water at  $20^\circ\text{C}$ . What mass of sucrose crystals will form if the temperature is reduced to  $10^\circ\text{C}$ ?

## Uses of solubility curves in the workplace

Sometimes it is important to know how much of a solute will precipitate out of a solution as it cools to a particular temperature, or if all the solute will stay dissolved. Two examples, one from industry and one from biological research, illustrate how solubility curves can be used to help in situations like this.

### Example 1: Crystallising sugar from sugar cane syrup

In Australia, table sugar (sucrose) is prepared in large quantities from sugar cane. During processing, the cane is crushed to release juice. Impurities are removed and water is evaporated from the juice until the solution becomes saturated and the sugar begins to crystallise out. Chemists use solubility curves like the one in Figure 15.6.10 to predict the amount of sucrose that will crystallise at different temperatures and solubilities.

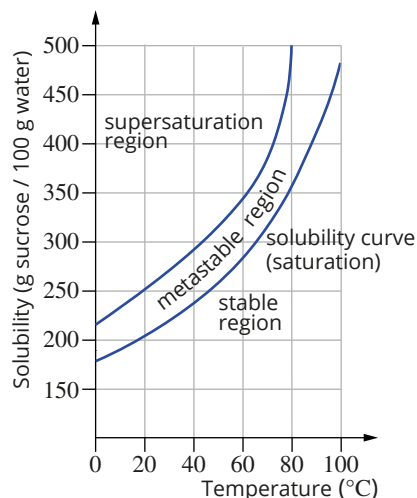
In this solubility curve, the area marked 'supersaturation region' indicates the conditions under which crystals will form spontaneously. In the 'metastable region', seed crystals must be added to induce crystallisation. In areas below the saturation curve, the solution is unsaturated and crystals will not form.

### Example 2: Research on therapeutic proteins

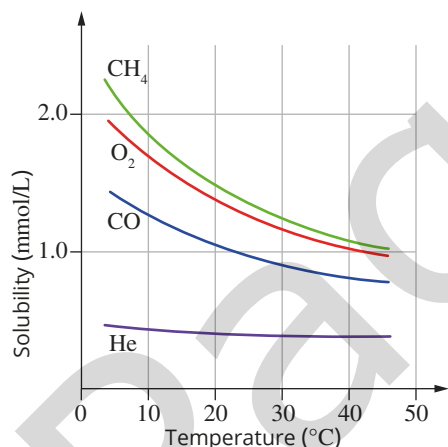
Therapeutic proteins are compounds that are manufactured in the laboratory for use as pharmaceuticals. Synthetic insulin is an example of a therapeutic protein. Therapeutic proteins need to remain dissolved when injected into the body.

Most biological reactions and many industrial processes take place in solution. Laboratory workers in medical, pharmaceutical and industrial fields need to know the temperature ranges at which compounds remain dissolved. If the temperature of the solutions they are using does not stay within a certain range, then reactants may crystallise out and interfere with the desired reactions.

Solubility curves allow researchers to make accurate predictions about the solubility at different temperatures of the compounds under investigation. They can then ensure that work will not be disrupted by unexpected and unwanted crystallisation.



**FIGURE 15.6.10** A solubility curve for sucrose used to predict the amount of sugar that will crystallise out under different conditions



**FIGURE 15.6.11** Solubility curves for some gases

## SOLUBILITY OF LIQUIDS AND GASES IN WATER

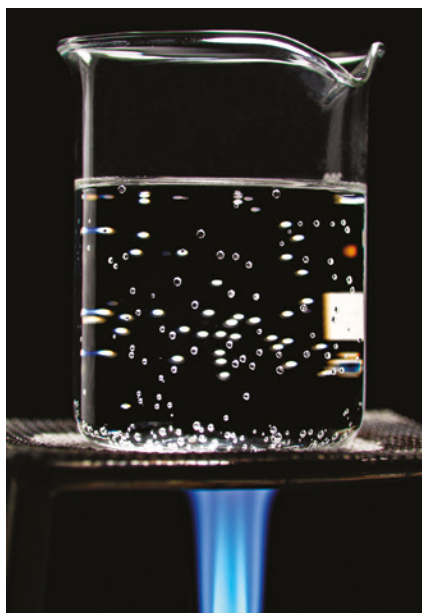
It is generally true that as temperature increases, the solubility of solids in solution increases. However, the solubility of liquids in other liquids does not show clear trends in solubility as the temperature changes.

Gases generally become less soluble as the temperature increases. The graph in Figure 15.6.11 shows the solubility of different gases in water at varying temperatures.

Table 15.6.2 shows the solubility of some common gases in water at different temperatures. For each gas, it can be seen that as temperature increases, less gas is able to dissolve in water.

**TABLE 15.6.2** Solubility of some gases in water at different temperatures

| Gas            | Solubility (g of gas per kg of water) at 0°C | Solubility (g of gas per kg of water) at 20°C | Solubility (g of gas per kg of water) at 60°C |
|----------------|----------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| oxygen         | 0.069                                        | 0.043                                         | 0.023                                         |
| carbon dioxide | 3.4                                          | 1.7                                           | 0.58                                          |
| nitrogen       | 0.029                                        | 0.019                                         | 0.011                                         |
| methane        | 0.040                                        | 0.023                                         | 0.011                                         |
| ammonia        | 897                                          | 529                                           | 168                                           |



**FIGURE 15.6.12** Bubbles of air appear as water is heated. This is because solubility of gases decreases at higher temperatures.



**FIGURE 15.6.13** Soft drinks left standing in the sun go 'flat' quickly.

In Figure 15.6.12, you can see the bubbles that appear when water is first heated are not steam (water vapour) but air bubbles. Some of the air that was dissolved in the water comes out of solution as the temperature increases and the gas becomes less soluble.

You may have noticed that soft drinks will go 'flat' more quickly if they are left standing in the sun on a hot day than if they are in a cold refrigerator (Figure 15.6.13). The dissolved carbon dioxide in the drink comes out of solution as the drink heats up.

The effect of temperature on the solubility of a gas can have environmental implications. If the temperature of the water in rivers, lakes and oceans increases even slightly, it will contain less dissolved oxygen. This can have serious consequences for oxygen-breathing aquatic organisms, such as the Australian Murray River cod in Figure 15.6.14.



**FIGURE 15.6.14** Fish, such as this native Australian Murray River cod, are susceptible to lowered levels of oxygen in natural waters.

Therefore, hot water from power stations and other industries must be cooled before it can be discharged into waterways. Even a small increase in water temperature can cause the oxygen concentration in the water to drop below levels that are necessary for aquatic life to survive.

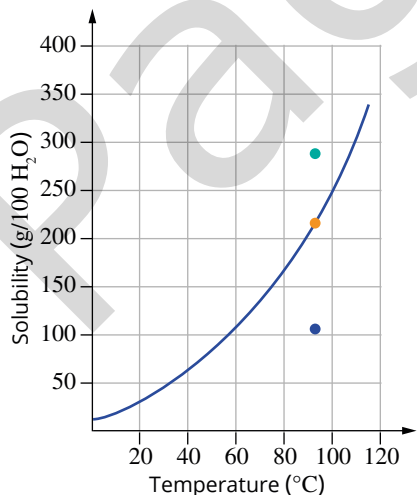
## 15.6 Review

### SUMMARY

- Solubility is a measure of how much solute will dissolve in a given amount of solvent at a specified temperature.
- Solubility curves show how the solubility of a compound changes with temperature.
- Solubility curves can be used to calculate the amount of a substance that will dissolve in a given amount of solvent at a specified temperature.
- The solubility of most solids in water increases as the temperature increases.
- Crystallisation occurs when a solution is cooled to the point where it becomes saturated with solute and the solute begins to form solid crystals.
- As the temperature of a solution increases, the solubility of solids generally increases and the solubility of gases decreases. Liquids show no overall trend in solubility with temperature.
- Crystallisation can occur from a solution when:
  - the temperature of the solution decreases
  - solvent evaporates.
- Factors that can affect crystallisation, including crystal size, are:
  - rate of cooling of solution
  - rate of evaporation of solvent
  - nucleation
  - nature of the compound.
- Solubility curves can be used to predict the amount of a compound that will precipitate from solution at a particular temperature.

### KEY QUESTIONS

- 1 35g of sodium chloride will dissolve in 100g of water at 18°C to form a saturated solution. A number of different solutions were made by dissolving the following masses of sodium chloride in 50g of water at 18°C. Which solution would be supersaturated?  
**A** 9g  
**B** 3.5g  
**C** 20g  
**D** 17.5g
- 2 Look at graph below and determine which coloured points correspond to saturated, unsaturated or supersaturated solutions of potassium nitrate.



- Refer to Figure 15.6.2 on page XX to answer Questions 3–7.
- 3 Find the mass of solute that could dissolve in the following situations.  
**a**  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$  in 100g of water at 60°C  
**b**  $\text{AgNO}_3$  in 100g of water at 10°C  
**c**  $\text{KNO}_3$  in 50g of water at 24°C
  - 4 What mass of the following compounds will dissolve in 200g of water at 30°C to form a saturated solution?  
**a**  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$   
**b**  $\text{NaNO}_3$   
**c**  $\text{NaCl}$
  - 5 140g of potassium nitrate ( $\text{KNO}_3$ ) is dissolved in 100g of water at 70°C. The temperature of the solution is then cooled to 55°C.  
What mass of  $\text{KNO}_3$  will crystallise out of solution?  
**A** 33g  
**B** 40g  
**C** 100g  
**D** 140g
  - 6 Calculate the mass of crystals that will form when a solution containing 60g of  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$  in 100g of water at 90°C is cooled to 60°C.
  - 7 55g of sodium nitrate was dissolved in 50g of water at 80°C. What mass of sodium nitrate will remain dissolved if the solution was cooled to 50°C?
  - 8 Explain why granite crystals are a different size from basalt crystals.



## Chapter review

### KEY TERMS

aqueous  
artesian basin  
bore water  
chemical contaminant  
chlorination  
crystallisation  
crystallise  
desalinated seawater  
desalination  
dipole  
dissociation  
dissolution  
floc  
flocculation

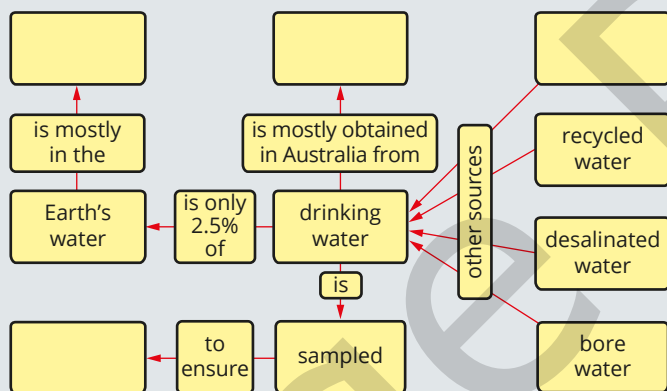
fluoridation  
Great Artesian Basin  
groundwater  
heat capacity  
heavy metal  
homogeneous  
hydrated  
hydride  
hydronium ion  
ion–dipole attraction  
ionise  
latent heat  
latent heat of fusion  
latent heat of vaporisation

miscible  
non-polar solvent  
osmosis  
polar  
polar solvent  
potable water  
protected catchment  
recycled water  
reverse osmosis  
saturated solution  
seed crystal  
settling  
solubility  
solubility curve

solubility table  
solute  
solution  
specific heat capacity  
sterile  
supersaturated solution  
unsaturated solution  
water cycle

### Essential water

- 1 Copy and complete the following diagram relating to drinking water and water quality.



- 2 Where is most of the drinkable water found on Earth?  
**A** ice  
**B** rivers  
**C** groundwater  
**D** bore water
- 3 What is the drinking water that comes from groundwater called?
- 4 What is the source of drinking water for most Australians and why?
- 5 Arrange the following sources of water on Earth in order from highest volume to lowest volume: oceans, groundwater, rivers, soil moisture, ground ice and permafrost, lakes, ice caps and glaciers, atmosphere as water vapour.

- 6 Imagine that you are an environmental scientist and have been asked to develop a water-sampling program for a large reservoir. The quality of the water is to be tested over a month. What are some of the matters that should be addressed before starting your program?
- 7 Which one of the following factors does not need to be considered when sampling water for analysis?  
**A** temperature  
**B** depth  
**C** distance from the shore  
**D** humidity
- 8 A water sample was tested for heavy metals. The results are shown in the table below. Classify which of the concentrations are above and which are below the Australian Drinking Water Guidelines acceptable levels (Table 15.1.5 on page XX).

| Heavy metal | Concentration (mg L <sup>-1</sup> ) |
|-------------|-------------------------------------|
| arsenic     | 0.009                               |
| cadmium     | $2.2 \times 10^{-4}$                |
| copper      | 1.9                                 |
| lead        | 0.1                                 |
| mercury     | $0.9 \times 10^{-2}$                |

- 9 Give a possible source of error in each of the following steps in sampling water from a lake.  
**a** determine sampling location  
**b** collect sample  
**c** transfer sample to a container  
**d** store sample  
**e** transport sample

### Properties of water

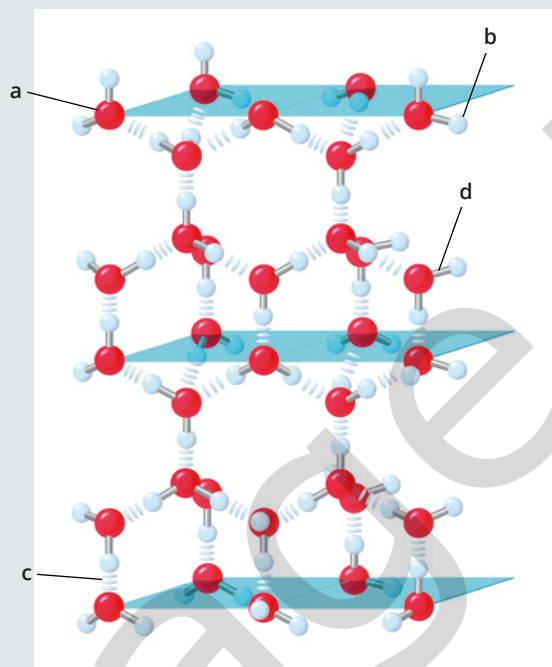
- 10** Copy and complete the following sentences about water and its properties.

Water is a \_\_\_\_\_ molecule. Within a single molecule, hydrogen and oxygen atoms are held together by strong \_\_\_\_\_. Between different molecules, the most significant forces are \_\_\_\_\_.

It is the relatively \_\_\_\_\_ strength of the intermolecular forces that gives water its unique properties of relatively:

- relatively \_\_\_\_\_ boiling point, \_\_\_\_\_ °C
- relatively \_\_\_\_\_ latent heat values, \_\_\_\_\_ kJ mol<sup>-1</sup> and \_\_\_\_\_ kJ mol<sup>-1</sup>
- relatively \_\_\_\_\_ specific heat capacity, \_\_\_\_\_ J g<sup>-1</sup> °C<sup>-1</sup>.

- 11** The image below shows the arrangement of water molecules in ice. Identify the parts of the diagram labelled **a–d**.



- 12** Water boils at 100°C. However, a much higher temperature (1000°C) is needed to decompose water molecules into hydrogen gas and oxygen gas.

- a** Using water as an example, explain the meaning of the terms ‘intermolecular’ and ‘intramolecular’ forces.
- b** Which of the two types of forces described in part **a** is stronger? Justify your answer by using the information at the beginning of the question.

- 13** Hydrogen sulfide and water are both group 16 hydrides. Explain why water exhibits a much higher boiling point than hydrogen sulfide.

- 14** In your own words, discuss the significance of polarity and hydrogen bonding in relation to the high boiling point of water.

- 15** An evaporative air conditioner is used to cool the air inside some buildings. The cooler lowers the air temperature by using heat from the air to evaporate water. Select the property that is most important when water is used in an evaporative air conditioner.

- 16** When heat energy is applied to a substance, the temperature increase of the substance depends on which of the following?

- A** the mass of the substance  
**B** the types of bonds present in the substance  
**C** the amount of heat energy transferred to the substance  
**D** all of the above

### Water as a solvent

- 17** Many of the solvent properties of water depend on its structure and bonding. List all the bonds present in a sample of water from weakest to strongest. Remember to include all intramolecular and intermolecular bonding.

- 18** Explain why water is such a good solvent for polar and ionic substances.

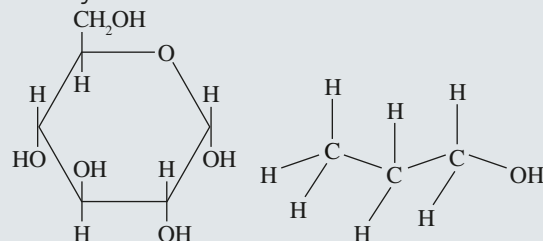
- 19** Nitrogen gas (N<sub>2</sub>) and ethene (C<sub>2</sub>H<sub>6</sub>) are both insoluble in water, whereas ethanol (C<sub>2</sub>H<sub>5</sub>OH) is soluble. Refer to the bonding in each of the molecules to explain these differences.

- 20** Explain, with reference to its structure and bonding, why octane does not dissolve in water.

### Water as a solvent of molecular substances

- 21** Classify the following substances according to the way they dissolve in water: C<sub>6</sub>H<sub>12</sub>O<sub>6</sub>, HI, I<sub>2</sub>, C<sub>2</sub>H<sub>4</sub>, C<sub>3</sub>H<sub>7</sub>OH, HNO<sub>3</sub>, CH<sub>4</sub>.

The structures of C<sub>6</sub>H<sub>12</sub>O<sub>6</sub> and C<sub>3</sub>H<sub>7</sub>OH are shown to assist you.



- 22** A student conducts an experiment to investigate the solubility of CH<sub>4</sub> and CH<sub>3</sub>OH molecules in three different solvents, X, Y and Z. The student finds that the CH<sub>3</sub>OH molecules dissolve well in solvent Y but do not dissolve in solvent X. The student also finds that CH<sub>4</sub> partially dissolves in solvent Z and completely dissolves in solvent X. List each of the solvents in order from most polar to least polar. Explain your reasoning.

- 23 DDT is a hazardous agricultural insecticide, which has been banned in many countries. It is only slightly soluble in water but is very soluble in fats and oils, so accumulates in the fat deposits of animals. What can you deduce about the polarity of the DDT molecule from its solubility characteristics?

### Water as a solvent of ionic compounds

- 24 a What is the name given to the process that ionic solids undergo when dissolving in water?  
b What ions will be produced when the following compounds are added to water?  
i  $\text{Cu}(\text{NO}_3)_2$   
ii  $\text{ZnSO}_4$   
iii  $(\text{NH}_4)_3\text{PO}_4$
- 25 What ions would be produced when the following compounds are added to water?  
a potassium carbonate  
b lead(II) nitrate  
c sodium hydroxide  
d sodium sulfate  
e magnesium chloride  
f zinc nitrate  
g potassium sulfide  
h iron(III) nitrate
- 26 Write equations to show the dissociation of the following compounds when they are added to water.  
a magnesium sulfate  
b sodium sulfide  
c potassium hydroxide  
d copper(II) ethanoate  
e lithium sulfate
- 27 What particles are present in the solution that forms when HCl gas is bubbled through water?
- 28 Briefly explain why the arrangement of water molecules around dissolved magnesium ions is different from that around dissolved chloride ions.
- 29 Write the formulae of three carbonate compounds that are:  
a soluble in water  
b insoluble in water.
- 30 Write the formulae of three sulfate compounds that are:  
a soluble in water  
b insoluble in water.
- 31 Briefly describe what happens to the forces between solute and solvent substances when an ionic substance such as potassium bromide dissolves in water.

### Solubility

- 32 What mass of the following compounds will dissolve in 50g of water at 30°C to form a saturated solution?  
a  $\text{NaNO}_3$   
b  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$   
c  $\text{KNO}_3$

- 33 What is the maximum mass of solute that will dissolve in the following situations?  
a sucrose in 500g of water at 20°C  
b  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$  in 40g of water at 60°C
- 34 160g of  $\text{KNO}_3$  was dissolved in 200g water at 60°C. What mass of crystals would form if the temperature were decreased to 30°C?
- 35 What effect does increasing the temperature have on the concentration of gases available to aquatic animals and plants?
- 36 A gas forms a saturated solution when 0.50mmol dissolves in 500mL water at 23°C. Use Figure 15.6.11 on page XXX to decide what the gas is most likely to be.
- 37 If 2.0kg water absorbed 1200g of ammonia at 10°C, what mass of the gas would leave the solution if the temperature of the solution was increased to 20°C? (Refer to Table 15.6.2 on page XXX.)

### Connecting the main ideas

- 38 Match the type of compound with the way it is likely to behave in water.
- i ionic compound
  - ii compound composed of polar molecules with –OH groups
  - iii compound composed of small polar molecules in which a hydrogen atom is covalently bonded to an atom of a group 17 element
  - iv non-polar molecular compound
  - v compound composed of covalent molecules with a large non-polar end and one –OH group.
- a does not dissolve in water because a large proportion of the molecule is non-polar  
b dissolves in water by ionising, then forming ion–dipole bonds with water  
c does not dissolve in water  
d dissolves in water by forming hydrogen bonds with water molecules  
e dissolves in water by dissociating, then forming ion–dipole bonds with water
- 39 Give concise explanations for the following observations.
- a Ammonia ( $\text{NH}_3$ ) and methane ( $\text{CH}_4$ ) are both covalent molecular substances. Ammonia is highly soluble in water, but methane is not.
  - b Glucose ( $\text{C}_6\text{H}_{12}\text{O}_6$ ) and common salt ( $\text{NaCl}$ ) are very different compounds. Glucose is a covalent molecular substance, whereas common salt is ionic, yet both of these substances are highly soluble in water.



